

First Exam Spring 2019/2020

09 July 2020 00:52

1. The y-intercept of the inverse of the function below is equal to
(1 Point)

$$f(x) = \frac{4x - 3}{3x + 2}$$

- ☐ $-\frac{3}{2}$
- ☒ $\frac{3}{4}$ ✓
- ☐ -1
- ☐ 2

2. Let $y=f(x)$ be the function defined by
(1 Point)

$$f(x) = x^3 + 2x + 2e^{3x-3}. \text{ Then } f^{-1}(5) \text{ is equal to}$$

- ☐ 0
- ☐ 2
- ☐ -1
- ☒ 1 ✓

3. The simplest form of
(1 Point)

$$\sin(2 \sin^{-1} x)$$

☒ $2x\sqrt{1-x^2}$ ✓

☐ $\frac{2x}{\sqrt{1-x^2}}$

☐ $\sin 2 \sin^{-1} x$

☐ $1 - 2x^2$

4. The range of the function $y = f(x)$, where
(1 Point)

$$f(x) = 2 \sin^{-1}(3x) + 5$$

☐ $[-1, 1]$

☐ $[3, 7]$

☐ $(-\infty, \infty)$

☒ $[-\pi + 5, \pi + 5]$ ✓

5. The domain of the function $y = f(x)$, where
(1 Point)

$$f(x) = \frac{\ln(x-2)}{e^{x-1} - 1} \text{ is}$$

☒ $(2, \infty)$ ✓

☐ $(-\infty, \infty) \setminus \{1\}$

☐ $(1, \infty)$

☐ $(0, \infty) \setminus \{1\}$

6. Suppose that the limit of the functions f and g exist when x approaches a .
(1 Point)

$$\text{If } \lim_{x \rightarrow a} (f(x) + 2g(x)) = 2, \lim_{x \rightarrow a} (-3f(x) + g(x)) = 5, \text{ then } \lim_{x \rightarrow a} (f(x)g(x)) =$$

☐ 10

☐ $-\frac{23}{35}$

☒ $-\frac{88}{49}$ ✓

☐ $\frac{42}{77}$

7. The point(s) of discontinuity of the given function is (are) at $x =$
(1 Point)

$$f(x) = \begin{cases} x^2, & \text{if } x < 0, \\ \ln(x+1), & \text{if } 0 \leq x \leq 1 \\ \frac{1}{x}, & \text{if } 1 < x \leq 2. \end{cases}$$

- ☐ 0
- ☒ 1 ✓
- ☐ 2
- ☐ 0, 1

8. The value of
(1 Point)

$$\lim_{x \rightarrow 0} \frac{|3x - 2| - |3x + 2|}{x} \text{ if it does exist is equal to}$$

- ☐ 0
- ☐ Does not exist
- ☒ -6 ✓
- ☐ 4

9. The value of
(1 Point)

$$\lim_{h \rightarrow 0} \frac{f(5+h) - f(5) + 2h - h^2}{3h} \text{ provided that } f'(5) = 4 \text{ is equal to}$$

- ☐ 0
- ☐ 1
- ☐ -1
- ☒ 2 ✓

10. The value of
(1 Point)

$\lim_{x \rightarrow \frac{1}{2}} f(x)$ provided that $4x \leq f(x) \leq 16x^4 - 4x^2 + 2$ is equal to

- ☐ 0
- ☐ 4
- ☐ -1
- ☒ 2 ✓

11. The given function is continuous on
(1 Point)

$$f(x) = \ln(x^2 + 1)$$

- ☐ $\mathbb{R} \setminus \{\pm 1\}$
- ☐ $(0, \infty)$
- ☐ $(-\infty, 0)$
- ☒ $\mathbb{R}$ ✓

12. If the functions f and g are continuous functions such that
(1 Point)

$g(1) = 3$ and $\lim_{x \rightarrow 1} (f(x) + 2f(x)g(x)) = -7$, then $f(1)$ is equal to

- ☒ -1 ✓
- ☐ 0
- ☐ -2
- ☐ 1

13. The values of the constant c that makes the given function continuous everywhere are
(1 Point)

$$f(x) = \begin{cases} cx - \ln(x-1), & \text{if } x > 2, \\ c^2x^2 - \frac{3}{2}cx - 2, & \text{if } x \leq 2. \end{cases}$$

- ☐ 0, 1
- ☐ $\pm \frac{1}{2}$
- ☒ $\frac{5 \pm \sqrt{57}}{8}$ ✓
- ☐ $\frac{3 \pm \sqrt{45}}{10}$

14. If $y=f(x)$ is a continuous function on all real numbers with $f(3) = -2$, $f(x)>0$ for $x > 4$ or $x < 0$, then one of the following statements must be true
(1 Point)

- ☐ f is differentiable
- ☒ f has at least two real roots ✓
- ☐ f has exactly three real roots
- ☐ f has at most two real roots

15. The horizontal asymptote (if any) of the graph of the given function is
(1 Point)

$$f(x) = \sqrt{x^2 + 3} + x$$

- ☐ $y = 1$
- ☐ $y = -1$
- ☒ $y = 0$ ✓
- ☐ The graph of f has no horizontal asymptotes

16. The value of the given limit if it exists is equal to
(1 Point)

$$\lim_{x \rightarrow \infty} \ln(5x^3 + 2x - 1) - \ln(2x^3 - x + 3)$$

- ☐ 0
- ☒ $\ln\left(\frac{5}{2}\right)$ ✓
- ☐ Does not exist
- ☐ 1

17. The value of the given limit if it does exist is equal to
(1 Point)

$$\lim_{x \rightarrow 1^+} \tan^{-1}\left(\frac{1}{x-1}\right)$$

- ☒ $\frac{\pi}{2}$ ✓
- ☐ 0
- ☐ 1
- ☐ Does not exist

18. The slope of the normal line to the graph of the given function at $x=0$ is
(1 Point)

$$f(x) = x^3 - xe^x + 3x - 1$$

- ☒ $-\frac{1}{2}$ ✓
- ☐ 0
- ☐ -2
- ☐ $\frac{3}{2}$

19. An equation of the tangent line to the graph of the given function at $x=0$ is
(1 Point)

$$f(x) = x^3 - 2xe^x + 3$$

- ☒ $y - 3 = -2(x - 0)$ ✓
- ☐ $y = -2x$
- ☐ $y - 3 = -2(x - 3)$
- ☐ $y = 3x$

20. The points where the tangent line to the graph of the given function is horizontal are
(1 Point)

$$f(x) = x^3 + 3x^2 - 9x + 4$$

- ☐ (0, 4), (1, -1)
- ☐ (-1, 15), (0, 4)
- ☐ (2, 6), (-3, 31)
- ☒ (1, -1), (-3, 31) ✓

21. The given function below has a vertical asymptote(s) at
(1 Point)

$$f(x) = \frac{x^2 - 3x + 2}{x^2 + 3x - 4}$$

- ☐ $x = 1$
- ☐ $x = 1, -4$
- ☒ $x = -4$ ✓
- ☐ $x = 4$

22. The value of the given limit if it does exist is equal to

(1 Point)

$$\lim_{x \rightarrow 1} \frac{2 - \sqrt{x+3}}{x-1}$$

☐ $-\frac{1}{4}$ ✓

☐ $\frac{1}{4}$

☐ 0

☐ -1

23. Assume that f is a continuous function at $x=-1$ and $f(-1) = 2$, then the value of the given limit if it does exist is equal to

(1 Point)

$$\lim_{x \rightarrow -1} (x-1)f(x)$$

☐ 0

☐ -1

☐ 1

☐ -4 ✓

24. If $f(2) = -3$ and $f(5) = 1$, then $f(x) = 0$ has at least one solution in the interval

(1 Point)

☐ (2, 5)

☐ (-3, 2)

☐ (-1, 3)

☐ None of these given choices ✓

25. The derivative of the given function f at $x=0$ is equal to

(1 Point)

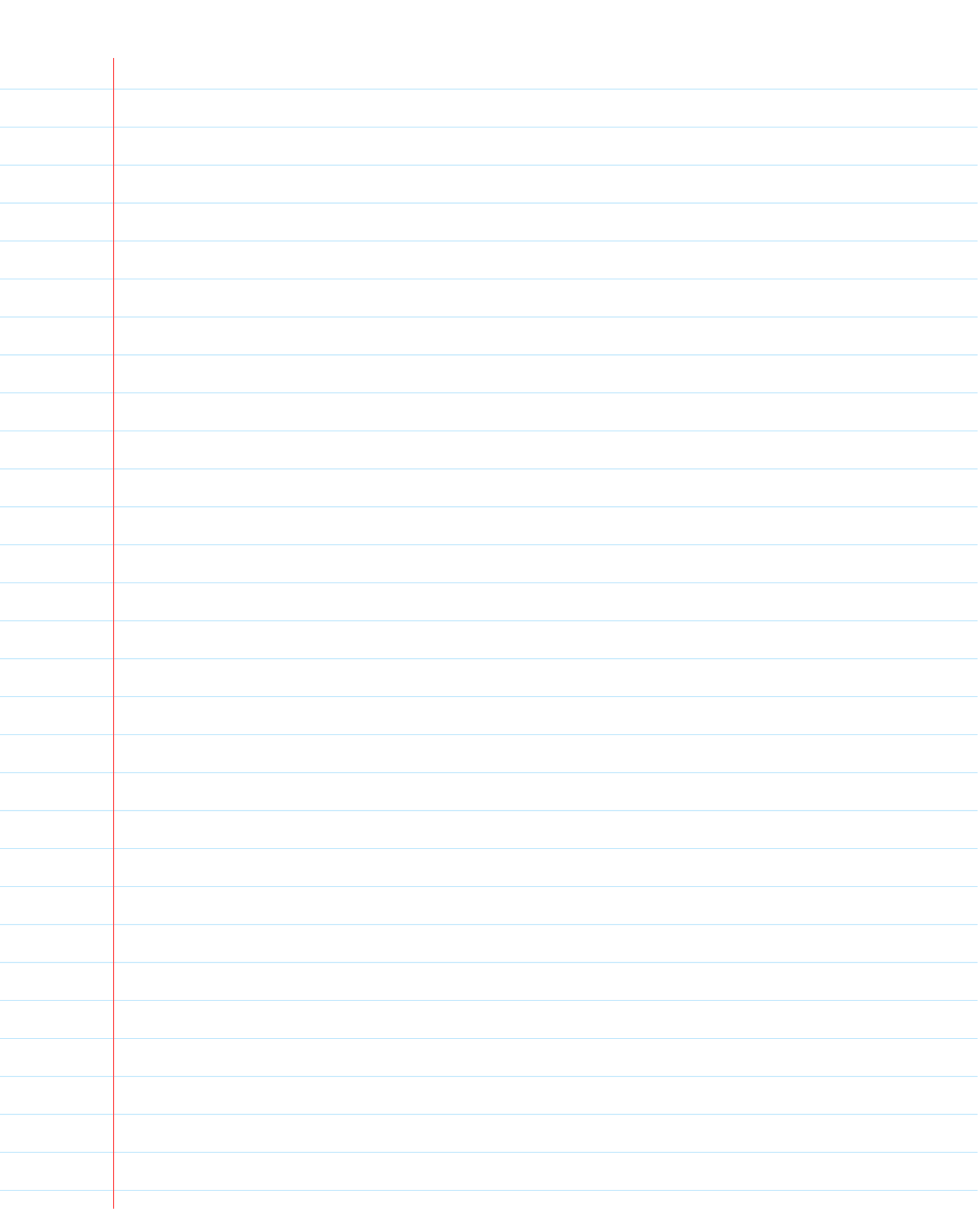
$$f(x) = \frac{xe^x}{x^2 - \sqrt{x^3} + 1}$$

☐ 1 ✓

☐ 0

☐ -1

☐ -2



Incomplete First Exam Spring2019/2020

20 October 2020 09:59

1. The y-intercept of the inverse of the function below is equal to
(1 Point)

$$f(x) = \ln|x| + \ln|x-1| \text{ for } x < 0$$

- ☐ $-e$
- ☒ $\frac{1-\sqrt{5}}{2}$ ✓
- ☐ -1
- ☐ $\frac{1+\sqrt{7}}{2}$
- ☐ None of these given choices

2. The simplest form of
(1 Point)

$$\cos(2 \sin^{-1} 3x)$$

- ☐ $6x\sqrt{1-9x^2}$
- ☐ $\frac{2x}{\sqrt{1-x^2}}$
- ☐ $1-2x^2$
- ☒ $1-18x^2$ ✓
- ☐ $1-38x^2$

3. The range of the function $y = f(x)$, where
(1 Point)

$$f(x) = \begin{cases} \sin^{-1}(2x), & \text{if } -\frac{1}{2} \leq x < 0, \\ \ln(x+1), & \text{if } 0 \leq x \end{cases}$$

- ☐ $[-1, 1]$
- ☐ $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- ☐ $(-\infty, \infty)$
- ☒ $\left[-\frac{\pi}{2}, \infty\right]$ ✓
- ☐ None of these given choices

4. The domain of the function $y = f(x)$, where
(1 Point)

$$f(x) = \frac{\ln(x+1)}{e^{2x} - 3e^x + 2} \text{ is}$$

- ☒ $(-1, \infty) \setminus \{0, \ln 2\}$ ✓
- ☐ $(-\infty, \infty) \setminus \{0\}$
- ☐ $(-1, \infty)$
- ☐ $(0, \infty) \setminus \{0\}$
- ☐ None of these given choices

5. Suppose that the limit of the functions f and g exist when x approaches a .
(1 Point)

$$\text{If } \lim_{x \rightarrow a} (f(x) + 2g(x)) = 6 \text{ and } \lim_{x \rightarrow a} (-3f(x) + g(x)) = 2, \text{ then } \lim_{x \rightarrow a} (3f(x) + g(x)) \text{ is equal to}$$

- ☐ 0
- ☐ $-\frac{43}{35}$
- ☒ $\frac{26}{7}$ ✓
- ☐ $\frac{23}{17}$
- ☐ None of these given choices

6. The given function is discontinuous at $x =$
(1 Point)

$$f(x) = \begin{cases} \frac{x^2}{x+1}, & \text{if } x \leq 0, \\ \ln x, & \text{if } 0 \leq x \leq 1 \end{cases}$$

6. The given function is discontinuous at $x =$

(1 Point)

$$f(x) = \begin{cases} \frac{x^2}{x+1}, & \text{if } x \leq 0, \\ \ln x, & \text{if } 0 \leq x \leq 1 \\ \frac{1}{x^2-1}, & \text{if } 1 < x. \end{cases}$$

- ☐ 0
- ☐ None of the given choices ✓
- ☐ 0, -1
- ☐ 0, 1
- ☐ ± 1

7. The value of

(1 Point)

$\lim_{x \rightarrow \infty} \frac{2x^2 + 5x^3 \tan^{-1}(2x+3)}{\sqrt{4x^6 - 2x + 5}}$ is equal to

- ☐ 0
- ☐ $\frac{5\pi}{4}$ ✓
- ☐ $\frac{-\pi}{2}$
- ☐ $\frac{5}{2}$
- ☐ None of the given choices

8. If $y=f(x)$ is a continuous function on $[2,6]$ with $f(3) = -2$ and $f(5) = 3$ then one of the following statements must be true

(1 Point)

- ☐ f is differentiable
- ☐ f has at least one real roots in $[2, 3]$
- ☐ f has exactly one real root in $[3, 6]$
- ☐ f has at most two real roots
- ☐ f has at least one real root in $[2, 5]$ ✓

9. An equation of the tangent line to the graph of the given function at $x=0$ is
(1 Point)

$$f(x) = \frac{2xe^x + 3}{x^2 + 1}$$

- ☒ $y - 3 = 2(x - 0)$ ✓
- ☐ $y = 2x$
- ☐ $y - 3 = 2(x - 3)$
- ☐ $y = x$
- ☐ None of these given choices

10. The points where the normal line to the graph of the given function is vertical are
(1 Point)

$$f(x) = (x^2 + 3)e^x$$

- ☐ $(0, 3), (1, 4e)$
- ☐ $\left(-3, \frac{12}{e^3}\right)$
- ☐ $(2, 7e^2)$
- ☒ $\left(-1, \frac{4}{e}\right), \left(-2, \frac{7}{e^2}\right)$ ✓
- ☐ None of these given choices

11. The given function below has a vertical asymptote(s) at
(1 Point)

$$f(x) = \frac{(x-1)\ln x}{x^2 - 2x + 1}$$

- ☐ $x = 1$
- ☐ $x = 0$
- ☒ $x = 0, 1$ ✓
- ☐ $x = 2$
- ☐ None of these given choices

12. The value of the given limit if it does exist is equal to
(1 Point)

$$\lim_{x \rightarrow 1} \frac{2 - \sqrt{x+3}}{1 - \sqrt{2x-1}}$$

- ☐ $-\frac{1}{4}$
- ☒ $\frac{1}{4}$ ✓
- ☐ 0
- ☐ -1
- ☐ None of the given choices

13. The inverse of the given function is equal to
(1 Point)

$$f(x) = \ln(x + \sqrt{x^2 + 1})$$

- ☐ $e^{x - \sqrt{x^2 + 1}}$
- ☒ $\frac{e^x - e^{-x}}{2}$ ✓
- ☐ $e^{x - \sqrt{x^2 - 1}}$
- ☐ $\ln(x - \sqrt{x^2 - 1})$
- ☐ None of these given choices

14. Question
(1 Point)

If $\lim_{x \rightarrow 1^+} f(x) = -2$ and $\lim_{x \rightarrow 1^-} f(x) = 2$ then $\lim_{x \rightarrow 1} (f(x))^2$ is equal to

- ☐ -4
- ☒ 4 ✓
- ☐ 0
- ☐ Does not exist
- ☐ None of these given choices

15. (1 Point)

The value of $\lim_{h \rightarrow 0} \frac{f(5+2h) - f(5) + 4h - h^2}{3h}$ if it exists provided that $f'(5) = 4$ is equal to

- ☐ 0
- ☒ 4 ✓
- ☐ -1
- ☐ 2
- ☐ None of these given choices

The value of $\lim_{x \rightarrow 2^+} \left(\frac{2x}{|x-2|} - \frac{x+2}{x-2} \right)$ is equal to

- ☐ 0
- ☐ -1
- ☒ 1 ✓
- ☐ Does not exist
- ☐ None of these given choices

17. (1 Point)

The value of a such that $\lim_{x \rightarrow 2} \frac{x^2 + ax - 2}{x^2 - 4} = \frac{3}{4}$ is

- ☐ 1
- ☒ -1 ✓
- ☐ 0
- ☐ 2
- ☐ None of these given choices

18. The value of the number b that assures that the given function has no real roots on $[0,2]$ is (1 Point)

$$f(x) = 1 - 6x + b^x$$

- ☐ 2
- ☐ $\frac{1}{2}$
- ☐ 3
- ☒ 5 ✓
- ☐ None of these given choices

19. Question
(1 Point)

The value of $\lim_{x \rightarrow -\infty} \frac{|x|(2x+1)}{\sqrt{5x^4+3x-1}}$ is equal to

- ☐ 2
- ☐ $\frac{2}{\sqrt{5}}$
- ☒ $-\frac{2}{\sqrt{5}}$ ✓
- ☐ 0
- ☐ None of these given choices

20. The tangent line to the graph of the given function at the point (0,4) intersects the graph a second time at x=

(1 Point)

$$f(x) = x^4 + 2x^3 - x + 4$$

- ☐ 1
- ☒ 2 ✓
- ☐ 3
- ☐ -1
- ☐ None of these given choices

21. Question
(1 Point)

If $f(x) = \lim_{t \rightarrow x} \frac{e^{t+2} - e^{x+2}}{t - x}$, then $f(1)$ if it does exist is equal to

- ☐ e
- ☐ e^2
- ☒ e^3 ✓
- ☐ e^4
- ☐ None of the given choices

22. (1 Point)

The value of $\lim_{x \rightarrow 0^+} \tan^{-1}(2 \ln x)$ is equal to

- ☐ 0
- ☐ $-\pi$
- ☐ $\frac{\pi}{2}$
- ☒ $\frac{-\pi}{2}$ ✓
- ☐ None of the given choices

23. Question
(1 Point)

If $\lim_{x \rightarrow 1} f(x) = 0$ and $\lim_{x \rightarrow 1} g(x)$ does not exist then $\lim_{x \rightarrow 1} f(x)g(x)$ is equal to

- ☐ 0
- ☐ 1
- ☐ Does not exist
- ☐ None of the given choices
- ☒ Can not be determined ✓

24. Suppose f is differentiable function at $x=0$ with the given property. Then the value of $f(0)$ is equal to
(1 Point)

$|f(x)| \leq x^2$ for all x

- ☐ -1
- ☐ 4
- ☐ 2
- ☐ Does not exist
- ☒ None of the given choices ✓

25. The value of the maximum number a such that if
(1 Point)

$|x - 2| < a$, then $|3x - 6| < 0.6$ is

- ☒ 0.2 ✓
- ☐ 0.3
- ☐ 0.4
- ☐ 0.1
- ☐ None of the given choices

First Exam (Summer)2019_2020

02 November 2020

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1. The y-intercept of the inverse of the function below is equal to
(3 Points)

$$f(x) = \frac{5x - 2}{3x + 4}$$

☒ $\frac{2}{5}$ ✓

☐ $\frac{2}{4}$

☐ -1

☐ 2

2. The relationship between the Fahrenheit (F) and Celsius (C) temperature scales is given by the linear function
(3 Points)

$$F = \frac{9}{5}C + 32.$$
 Which of the following is true.

☒ F increases. The F-intercept of 32 is the Fahrenheit temperature corresponding to 0°C . ✓

☐ F increases. The F-intercept of $9/5$ is the Fahrenheit temperature corresponding to 0°C .

☐ F increases. The F-intercept of 0 is the Fahrenheit temperature corresponding to 0°C .

☐ F decreases. The F-intercept of 32 is the Fahrenheit temperature corresponding to 0°C .

3. (3 Points)

$$\text{If } f(x) = e^{\frac{x+2}{3}}, \text{ find } f^{-1}(e^3)$$

☒ $6 - e^3$ ✓

☐ $6 - e$

☐ $1 - e^3$

☐ 1

4. The exact value of the expression
(3 Points)

$\cot\left(\cos^{-1}\frac{\sqrt{3}}{2}\right)$ is equal to

- ☒ $\sqrt{3}$ ✓
- ☐ 1
- ☐ $\sqrt{4}$
- ☐ $\sqrt{5}$

5. Find the domain and range of the function
(4 Points)

$$f(x) = 2 \sin^{-1}(3x + 1) + \pi$$

- ☒ Domain(f) = $[-\frac{2}{3}, 0]$ and Range(f) = $[0, 2\pi]$ ✓
- ☐ Domain(f) = $[-2, 2]$ and Range(f) = $[0, 2\pi]$
- ☐ Domain(f) = $(-\infty, \infty)$ and Range(f) = $[-\pi, \pi]$
- ☐ Domain(f) = $[-\frac{2}{3}, 0]$ and Range(f) = $[-\pi, \pi]$

6. Find the domain of the function
(3 Points)

$$f(x) = \frac{\ln(e^x - 2)}{x^2 - 9}$$

- ☒ $(\ln 2, 3) \cup (3, \infty)$ ✓
- ☐ $(-\infty, \infty) \setminus \{\pm 3\}$
- ☐ $(\ln 2, \infty) \setminus \{-3\}$
- ☐ $[2, 3) \cup [3, 9)$

7. Which of the following statement is true?

(4 Points)

- ☐ If f is one-to-one, then $f^{-1}(x) = \frac{1}{f(x)}$
- ☐ f is one-to-one $\leftrightarrow$ no horizontal line intersects its graph more than once ✓
- ☐ If $x > 0$ and $a > 0$ then, $\frac{\ln x}{\ln a} = \ln \frac{x}{a}$
- ☐ $\cos^{-1}(\cos x) = x$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

8. Suppose that

(4 Points)

$f(x) = 2x - 1$, $g(x) = \frac{x}{2} - \frac{1}{x}$, then find $\lim_{x \rightarrow 2} \frac{f(g(x))}{x - 2}$

- ☐ $\frac{3}{2}$ ✓
- ☐ $\frac{-2}{3}$
- ☐ $\frac{3}{4}$
- ☐ $\frac{-4}{3}$

9. Which of the following is true about

(4 Points)

$$f(x) = \begin{cases} x + 1, & \text{if } x < 1, \\ \frac{1}{x}, & \text{if } 1 < x < 3 \\ \sqrt{x-3}, & \text{if } x \geq 3. \end{cases}$$

- ☐ f is discontinuous at $x = 1, 3$, but it is continuous from the right at 3 ✓
- ☐ f is discontinuous at $x = 1, 3$, but it is continuous from the left at 3
- ☐ f is discontinuous at $x = -1$, but it is continuous from the right at 3
- ☐ None of these given choices

10. The value of
(3 Points)

$\lim_{x \rightarrow 1^+} \frac{1 - x^2}{|1 - x|}$ if it does exist is equal to

- ☒ -2 ✓
- ☐ Does not exist
- ☐ -3
- ☐ 3

11. Given that
(3 Points)

$\lim_{x \rightarrow 2} (5x - 7) = 3$, using the ϵ, δ definition of a limit, find a value of δ that correspond to $\epsilon = 0.1$

- ☒ $\delta = 0.02$ ✓
- ☐ $\delta = 0.1$
- ☐ $\delta = 0.01$
- ☐ None of those

12. Which of the following statement is FALSE about
(4 Points)

$$f(x) = \begin{cases} e^{x-1}, & \text{if } x < 1, \\ \frac{1}{x}, & \text{if } 1 \leq x < 5 \\ 5 - x, & \text{if } x \geq 5. \end{cases}$$

- ☒ $\lim_{x \rightarrow 1^-} f(x) = 0$ ✓
- ☐ $\lim_{x \rightarrow 1} f(x) = 1$
- ☐ $\lim_{x \rightarrow 5^-} f(x) = \frac{1}{5}$
- ☐ $\lim_{x \rightarrow 5^+} f(x) = 0$

13. Evaluate
(3 Points)

$$\lim_{t \rightarrow \infty} \frac{1 - t^{3/2}}{5t\sqrt{t} + 2t - 3} = ?$$

☒ $-\frac{1}{5}$ ✓

☐ 0

☐ 5

☐ $-\frac{2}{5}$

14. Find the values of the constant b that makes the given function continuous everywhere
(4 Points)

$$f(x) = \begin{cases} \frac{1}{b+e^{1/x}}, & \text{if } x < 0, \\ x^2 + b, & \text{if } x \geq 0 \end{cases}$$

☒ ± 1 ✓

☐ $\pm \frac{1}{2}$

☐ ± 2

☐ $\pm e$

15. Which of the following statements is true:
(3 Points)

If $f(x) = x^4 - x^2 + 1$ defined on the interval $[-2, 2]$, then the function

☒ has no zero in the interval ✓

☐ has at least one zero in the interval

☐ has more than two zeros in the interval

☐ None of these given choices

16. Find the horizontal and vertical asymptotes of the graph of the function
(4 Points)

$$f(x) = \frac{5x^3}{\sqrt{x^6 - 1}}$$

- ☒ HA $y = \pm 5$, VA $x = \pm 1$ ✓
- ☐ HA $y = \pm 1$, VA $x = \pm 5$
- ☐ HA $y = 5$, VA $x = \pm \frac{1}{5}$
- ☐ The graph of f has no horizontal and vertical asymptotes

17. Which of the following statement is NOT true
(3 Points)

- ☒ $f(x) = |x|$ is differentiable at $x = 0$ ✓
- ☐ $f(x) = \frac{|x|}{x}$ is not differentiable at $x = 0$
- ☐ If f is not continuous at a , then f is not differentiable at a
- ☐ If the graph of a function f has a corner in it, then f is not differentiable there.

18. (3 Points)

If $f(x) = (xe^{2x} - \sqrt{x^5} - e^{-x})(x + e)$, find $f'(0)$

- ☒ $2e - 1$ ✓
- ☐ $e^2 + 1$
- ☐ $2e^2 - 1$
- ☐ $e(3e + 1)$

19. Find an equation of the tangent line to the curve
(3 Points)

$y = x\sqrt{x}$ that is parallel to the line $y = 1 + 3x$

☒ $y = 3x - 4$ ✓

☐ $y = \sqrt{3}x + 4$

☐ $y = 5x - 4$

☐ $y = x - 4$

20. The slope of the normal line to the graph of the given function at $x=0$ is
(4 Points)

$f(x) = x(\sin^2 x - \cos^2 x) + 3x - 4$

☒ $-\frac{1}{2}$ ✓

☐ 0

☐ -2

☐ $\frac{3}{2}$

21. The position function of a particle is given below. What is the acceleration after 2 seconds?
(3 Points)

$s = t^3 - 10.5t^2 - t, \quad t > 0$

☒ The acceleration decreases with $9m/s^2$ ✓

☐ The acceleration increases with $7m/s^2$

☐ The acceleration decreases with $8m/s^2$

☐ The acceleration increases with $9m/s^2$

22. Which of the following is the value of the second derivative of the given function at $x=1$
(4 Points)

$$f(x) = \frac{(x^2 + 1)^3}{x}$$

- ☒ 40 ✓
☐ 41
☐ 42
☐ 43

23. Which of the following statement is NOT true
(3 Points)

- ☐ The velocity at time $t = a$ of an equation of the motion $s = f(t)$ is equal to the slope of the tangent line at $(a, f(a))$
☒ $f'(a) = \frac{f(a+h) - f(a)}{h}$ if this exists ✓
☐ If $f(x) \leq g(x) \leq h(x)$ when x is near a except possible at a and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} g(x) = L$
☐ $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$

24. The point where the tangent line to the graph of the given function is horizontal is
(3 Points)

$$f(x) = x + 2 \cos x, \quad 0 \leq x \leq \frac{\pi}{2}$$

- ☒ $\left(\frac{\pi}{6}, \frac{\pi}{6} + \sqrt{3}\right)$ ✓
☐ $\left(\frac{\pi}{3}, \frac{\pi}{3} + \sqrt{3}\right)$
☐ $\left(\frac{\pi}{6}, \frac{\pi}{6}\right)$
☐ $\left(\frac{\pi}{6}, \sqrt{3}\right)$

25. Find the derivative of the function.

(4 Points)

$$f(x) = \tan(\cos \sqrt{7x})$$

☒ $-\frac{\sqrt{7} \sin \sqrt{7x} \sec^2(\cos \sqrt{7x})}{2\sqrt{x}}$ ✓

☐ $-\frac{\sqrt{7} \cos \sqrt{7x} \sec^2(\cos \sqrt{7x})}{2\sqrt{x}}$

☐ $-\frac{\sqrt{7} \cos \sqrt{7x} \tan^2(\cos \sqrt{7x})}{2\sqrt{x}}$

☐ $\frac{\sqrt{7} \sin \sqrt{7x} \sec^2(\cos \sqrt{7x})}{2\sqrt{7x}}$

26. The value of the given limit if it does exist is equal to

(3 Points)

$$\lim_{x \rightarrow \frac{3}{2}} \frac{\sin(2x - 3)}{(2x - 3)(2x + 1)}$$

☐ $-\frac{1}{2}$

☒ $\frac{1}{4}$ ✓

☐ does not exist

☐ -1

27. Find an equation of the tangent line to the curve at the indicated point.

(3 Points)

$$y = x + \sin xy; \quad (0, 0)$$

☐ $y = 1$

☐ $x = \frac{\pi}{2}$

☒ $y = x$ ✓

☐ $y = x - 1$

28. Which of the following is the derivative of the given function
(3 Points)

$$f(x) = \frac{1}{2}(1 + 4x^2) \arctan 2x$$

- ☒ $f'(x) = 4x \arctan 2x + 1$ ✓
- ☐ $f'(x) = (1 + 4x^2) \arctan 2x + 2$
- ☐ $f'(x) = 4x^2 \arctan 2x - 1$
- ☐ None of these given choices

29. Find the rate of change of y with respect to x at the given values of x and y
(3 Points)

$$2xy^2 - 5x^2y + 192 = 0; \quad x = 4, y = 4$$

- ☒ -8 ✓
- ☐ -9
- ☐ -16
- ☐ None of these given choices

30. (3 Points)

$$\text{If } f(x) = \frac{x}{\ln x}, \text{ find } f'(e^4)$$

- ☒ $\frac{3}{16}$ ✓
- ☐ $\frac{3}{8}$
- ☐ $\frac{1}{8}$
- ☐ $-\frac{3}{8}$

Final Exam (Summer)2019_2020

02 November 2020

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1. Evaluate

(3 Points)

$$\lim_{x \rightarrow a^2} \frac{a^4 - x^2}{\sqrt{x} - a}, \text{ where } a \text{ is a constant and } a > 0$$

- ☒ $-4a^3$ ✓
- ☐ *Does not exist*
- ☐ $-3a^3$
- ☐ 0

2. Evaluate

(3 Points)

$$\lim_{t \rightarrow 0} \frac{\sin(\sinh t) + \tanh t}{e^t - e^{-t}}$$

- ☐ $\frac{1}{e}$
- ☒ 1 ✓
- ☐ ∞
- ☐ -1

3. If the constants a and b are the values that make the given function differentiable, (4 Points)

$$f(x) = \begin{cases} ax^3 & \text{if } x \leq 2 \\ x^2 + b & \text{if } x > 2 \end{cases}, \quad \text{then find } a + b$$

- ☒ $a + b = -1$ ✓
- ☐ $a + b = 1$
- ☐ $a + b = 2$
- ☐ $a + b = -2$

4. Which of the following is true about (3 Points)

$$f(x) = \arctan\left(\frac{x-1}{2-x}\right)$$

- ☒ f has a jump discontinuity at $x=2$ and has a horizontal asymptote at $y = \frac{-\pi}{4}$ ✓
- ☐ f has a jump discontinuity at $x=2$ and has no horizontal asymptote
- ☐ f has a jump discontinuity at $x=2$ and has a horizontal asymptote at $y = 2$
- ☐ f has an infinite discontinuity at $x=2$ and has a horizontal asymptote at $y = \frac{-\pi}{4}$

5. Which of the following statements is TRUE? (3 Points)

- ☒ $f(x) = x^{2/3}$ is continuous at 0, but it is not differentiable at 0 ✓
- ☐ If $f'(x) = g'(x)$ for $0 < x < 1$, then $f(x) = g(x)$ for $0 < x < 1$.
- ☐ $\frac{d}{dx}(10^x) = x10^{x-1}$
- ☐ If f has an absolute minimum value at c , then $f'(c) = 0$.

6. (4 Points)

$$\text{Find } \frac{d^4}{dx^4}(\ln(2x+1)) =$$

- ☒ $-2^4 \cdot 3!(2x+1)^{-4}$ ✓
- ☐ $2^5 \cdot 2!(2x+1)^{-4}$
- ☐ $2^4 \cdot 2!(2x+1)^{-4}$

6. (4 Points)

Find $\frac{d^4}{dx^4}(\ln(2x + 1)) =$

- ☒ $-2^4 \cdot 3!(2x + 1)^{-4}$ ✓
- ☐ $2^5 \cdot 2!(2x + 1)^{-4}$
- ☐ $2^4 \cdot 2!(2x + 1)^{-4}$
- ☐ $-2^5 \cdot 2!(2x + 1)^{-5}$

7. (4 Points)

The function $f(x) = x^3 + ax^2 + bx + 1$ has a local minimum at $x = -1$ and a local maximum at $x = -3$. The inflection point of $f(x)$ is :

- ☒ $(-2, -1)$ ✓
- ☐ $(-2, 2)$
- ☐ $(2, -1)$
- ☐ $(2, 1)$

8. Two cars, A and B, start moving from the same point. Car A travels south at 60 km/h and the other travels west at 25 km/h. At what rate is the distance between the cars increasing when car A is 8 km and car B is 6 km from the starting point?

(3 Points)

- ☒ 63 km/h ✓
- ☐ 61 km/h
- ☐ -60 km/h
- ☐ 55 km/h

9. (4 Points)

a) Find the linearization $L(x)$ of the given function f at 0,

b) Compute dy and Δy for the values of $x = \pi$ and $\Delta x = \frac{\pi}{4}$ for the given function,

$$f(x) = \sin x + 2 \cos x$$

☒ $L(x) = 2 + x$ and $dy = -\frac{\pi}{4}$, $\Delta y = \frac{4 - 3\sqrt{2}}{2}$ ✓

☐ $L(x) = 2 + x$ and $dy = \frac{\pi}{4}$, $\Delta y = \frac{4 - 3\sqrt{2}}{2}$

☐ $L(x) = 2 - x$ and $dy = -\frac{1}{4}$, $\Delta y = \frac{4 - 3\sqrt{2}}{2}$

☐ $L(x) = 2 - x$ and $dy = \frac{\pi}{4}$, $\Delta y = \frac{4 - 3\sqrt{3}}{2}$

10. By using the linear approximation of the given function
(3 Points)

$$f(x) = \sqrt{x+1} \text{ at } a = 0, \sqrt{1.1} \text{ is approximately equal to}$$

☒ $\frac{21}{20}$ ✓

☐ $\frac{25}{20}$

☐ $\frac{20}{21}$

☐ $\frac{23}{21}$

11. The slope of the normal line to the graph of the given function
(3 Points)

$$f(x) = \log_2(\operatorname{sech} x) \text{ at } x = \ln 2 \text{ is}$$

☒ $\frac{5 \ln 2}{3}$ ✓

☐ $-\frac{3}{5 \ln 2}$

☐ $\frac{3}{5 \ln 2}$

☐ $-\frac{3 \ln 2}{5}$

12. Find the vertical tangent(s), if any of the graph of the given function F
(3 Points)

$$F(x) = \int_0^x (3t - 2)^{-4/5} dt$$

- ☒ $\frac{2}{3}$ ✓
- ☐ $\frac{3}{2}$
- ☐ There is no vertical tangent line
- ☐ $\pm \frac{4}{5}$

13. Which of the following statements is true about
(4 Points)

$$f(x) = \begin{cases} 1 - x^2 & \text{if } 0 \leq x < 2, \\ 2x - 4, & \text{if } 2 \leq x \leq 3 \end{cases}$$

- ☒ Absolute maximum $f(3) = 2$; no local maximum. No absolute or local minimum ✓
- ☐ Absolute maximum $f(3) = 2$; no local maximum. Absolute and local minimum $f(0) = 1$
- ☐ Absolute maximum $f(3) = 2$; no local maximum. Absolute and local minimum $f(2) = -3$
- ☐ No absolute or local maximum . Absolute and local minimum $f(0) = 1$

14. The given function satisfies the hypotheses of the Mean Value Theorem on the interval $[0, 1]$.
Which of the following value(s) of c that satisfy the conclusion of the theorem.
(3 Points)

$$f(x) = x + e^x$$

- ☒ $\ln(e - 1)$ ✓
- ☐ $\ln(1 - e)$
- ☐ $e - 1$
- ☐ ± 1

15. Which of the following statements is NOT true:
(4 Points)

- ☐ If $f(1) = 7$ and $f'(x) \geq 4$ for $1 \leq x \leq 3$, then the smallest possible value of $f(3)$ is 15.
- ☐ If $f'(x)$ exists and nonzero for all x , then $f(1) \neq f(0)$.
- ☒ None of these ✓

16. Which of the following statements is true about

(3 Points)

$$f(x) = x - 4\sqrt{x}$$

- ☒ f is decreasing on $(0, 4)$ and increasing on $(4, \infty)$. Local minimum $f(4) = -4$. ✓
- ☐ f is decreasing on $(0, 4)$ and increasing on $(4, \infty)$. No local minimum.
- ☐ f is decreasing on $(0, \frac{1}{4})$ and increasing on $(\frac{1}{4}, \infty)$. Local minimum $f(4) = -4$.
- ☐ f is decreasing on $(0, \frac{1}{4})$ and increasing on $(\frac{1}{4}, \infty)$. No local minimum.

17. (3 Points)

How many real roots does the equation $x^5 - 7x + 3 = 0$ have in the interval $[-1, 1]$?

- ☒ at most one real root ✓
- ☐ at most five real roots
- ☐ at most two real roots
- ☐ no real roots

18. Which of the following statements is true about

(3 Points)

$$f(x) = x|x|.$$

- ☐ f has an inflection point at $(0,0)$ and $f''(0)$ exists
- ☐ f has no inflection point and $f''(0)$ exists
- ☒ f has an inflection point at $(0,0)$ and $f''(0)$ does not exist ✓
- ☐ f has no inflection point and $f''(0)$ does not exist

19. (4 Points)

A box with a square base and open top must have a volume of 4000 cm^3 . Find the dimensions of the box that minimize the amount of material used.

- ☒ $20 \times 20 \times 10$ ✓
- ☐ $5 \times 5 \times 160$
- ☐ $10 \times 10 \times 40$
- ☐ $20 \times 20 \times 5$

20. Find the point on the line $y=4x+8$ that is closest to the origin.

(3 Points)

- ☒ $(-\frac{32}{17}, \frac{8}{17})$ ✓
- ☐ $(-\frac{12}{17}, \frac{11}{17})$
- ☐ $(-\frac{32}{17}, \frac{8}{11})$
- ☐ $(-\frac{12}{17}, -\frac{2}{11})$

21. Which of the following statements is NOT true

(3 Points)

- ☐ $\frac{d}{dt} \int_0^1 e^{\arctan x} dx = 0$
- ☐ $\int_{-1}^1 \frac{\sin x}{1+x^2} dx = 0$
- ☒ $\int_{-4}^4 (x^5 - 2x^2 + 1) dx = 0$ ✓
- ☐ $3 \leq \int_1^4 \sqrt{x} dx \leq 6$

22. Find the most general antiderivative of the function

(3 Points)

$$f(x) = \sqrt[3]{x^2} + 2 \sinh x - \frac{1}{1+x^2}$$

- ☒ $F(x) = \frac{3}{5}x^{5/3} + 2 \cosh x - \arctan x + C$ ✓
- ☐ $F(x) = \frac{2}{3}x^{2/3} + 2 \cosh x - \arctan x + C$
- ☐ $F(x) = \frac{3}{5}x^{5/3} + 2 \sinh x - \arctan x + C$
- ☐ $F(x) = \frac{3}{5}x^{5/3} + 2 \cosh x - \arcsin x + C$

23. (3 Points)

Given that the graph of f passes through the point $(1, 6)$ and that the slope of its tangent line at $(x, f(x))$ is $2x + 1$, find $f(2)$.

- ☒ 10 ✓
- ☐ 9
- ☐ 5
- ☐ 7

24. (4 Points)

Let $\int_1^{x^2} (4\sqrt{t} - t^{3/2}) dt$ where $x \geq 0$, then the function is concave downward in the interval(s)

- ☒ $(\sqrt{2}, \infty)$ ✓
- ☐ $(0, \sqrt{2})$
- ☐ $(0, \infty)$
- ☐ $(0, \frac{1}{2}), (1, \infty)$

25. Suppose that $f(3)=7$, $f(1)=1$ and derivative of f is continuous. Find the value of
(3 Points)

$$I = \int_1^3 \frac{x \cdot f'(x) - f(x)}{x^2} dx$$

- ☒ $I = \frac{4}{3}$ ✓
- ☐ $I = \frac{1}{3}$
- ☐ $I = \frac{1}{4}$
- ☐ $I = \frac{3}{4}$

26. Find an equation of the tangent line to the given curve $y=F(x)$ at the point with x -coordinate 2
(4 Points)

$$F(x) = \int_2^x e^{(t^2-1)} dt$$

- ☒ $y = e^3(x - 2)$ ✓
- ☐ $y = e^3(x - 2) + 2$
- ☐ $y = e^4(x - 2) - 1$
- ☐ $y = e^{(x^2-1)}$

27. Rewrite the integral using the given substitution.

(3 Points)

$$\int \sin^3 x \sqrt{\cos x} \, dx, \quad u = \cos x$$

☒ $\int (u^{5/2} - u^{1/2}) \, du$ ✓

☐ $\int (u^{7/2} - u^{3/2}) \, du$

☐ $\int (u - 1) u^{3/2} \, du$

☐ $\int (u^{5/2} + 1) u^2 \, du$

28. Evaluate

(3 Points)

$$I = \int_1^2 \frac{\sec(\ln x) \cdot \tan(\ln x)}{x} \, dx,$$

☒ $I = \sec(\ln 2) - 1$ ✓

☐ $I = \tan(\ln 2) - 1$

☐ $I = \sec(\ln 2) - \tan(\ln 2)$

☐ $I = \ln 2 - 1$

29. The area of the region bounded by the curves

(4 Points)

$$y = 2x^2, \quad y = x^4 - 2x^2, \quad \text{is equal to}$$

☒ $\frac{128}{15}$ ✓

☐ $\frac{64}{15}$

☐ $\frac{32}{15}$

☐ $\frac{1}{15}$

30. Which of the following finds the area of the region bounded by the curves
(3 Points)

$y = \arcsin x$, $y = 1$ and $x = 0$,

☒ $\int_0^1 \sin y \, dy$ ✓

☐ $\int_0^1 \arcsin y \, dy$

☐ $\int_0^1 \sin x \, dx$

☐ $\int_0^{\sin 1} \sin x \, dx$

Sol of final exam (summer)2019_2020

19 November 2020

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15/11/2020. Final Exam

① $\lim_{x \rightarrow 2} \frac{d^2 - x^2}{\sqrt{x} - 2} = \lim_{x \rightarrow 2} \frac{(d^2 - x)(d^2 + x)}{\sqrt{x} - 2}$
 $= \lim_{x \rightarrow 2} \frac{(d^2 - x)(d^2 + x)(d^2 + x)}{\sqrt{x} - 2} = \frac{2d^2 \cdot 3d^2}{\sqrt{2} - 2} = -6d^3$

② $\lim_{t \rightarrow 0} \frac{\sin(\sinh t) + \tanh t}{e^t - e^{-t}}$
 $= \lim_{t \rightarrow 0} \frac{\sin(\sinh t) + \tanh t}{2 \sinh t} = \frac{1}{2} + \frac{1}{2} = 1$

③ $f(x) = \begin{cases} ax^2 & \text{if } x \leq 2 \\ 2x + b & \text{if } x > 2 \end{cases}$
 for diff: $3ax^2 = 2x$ at $x=2$
 $12a = 4 \Rightarrow a = \frac{1}{3}$
 for con: $\lim_{x \rightarrow 2^-} ax^2 = \lim_{x \rightarrow 2^+} 2x + b$
 $8a = 4 + b \Rightarrow b = 8 \cdot \frac{1}{3} - 4 = -\frac{4}{3}$
 $a + b = \frac{1}{3} - \frac{4}{3} = -1$

④ $f(x) = \arctan\left(\frac{x-1}{2-x}\right)$
 $\lim_{x \rightarrow 2^-} \arctan\left(\frac{x-1}{2-x}\right) = \arctan(-\infty) = -\frac{\pi}{2}$
 $\lim_{x \rightarrow 2^+} \arctan\left(\frac{x-1}{2-x}\right) = \arctan(\infty) = \frac{\pi}{2}$
 $\lim_{x \rightarrow 2} -\arctan\left(\frac{x-1}{2-x}\right) = -\arctan(\infty) = -\frac{\pi}{2}$

⑤ $f(x) = x^{2/3}$ is continuous at 0 but not diff. because $f'(x) = \frac{2}{3}x^{-1/3}$
 • Check: $f(x) = x^{2/3}$, $g(x) = x^{1/3}$
 $f = g^2$ but not $f(x) = g(x)^2$
 • $f(x) = |x|$ at $x=0$ absolute min but $f'(x)$ DNE at $x=0$ so $f'(0) \neq 0$
 • $\frac{d}{dt}(10^t) = 10^t \ln 10$

⑥ $\frac{d^4(C_0(2x+1))}{dx^4} = ?$
 $y = \frac{2}{(2x+1)} = 2(2x+1)^{-1}$
 $y' = 2(-1)(2x+1)^{-2} \cdot 2$
 $y'' = 2(-1)(-2)(2x+1)^{-3} \cdot 2$
 $y''' = 2(-1)(-2)(-3)(2x+1)^{-4} \cdot 2$
 $y^{(4)} = -2^4 \cdot 3! (2x+1)^{-4}$

⑦ $f(x) = x^3 + ax^2 + bx + 1$
 Local max $f'(-1) = 0$
 Local min $f'(2) = 0$
 $f'(x) = 3x^2 + 2ax + b$
 $f'(-1) = 3 - 2a + b = 0$
 $f'(2) = 12 + 4a + b = 0$
 $3 - 2a + b = 0$
 $12 + 4a + b = 0$
 $3 - 12 + 6a = 0 \Rightarrow 6a = 9 \Rightarrow a = \frac{3}{2}$
 $3 - 2(\frac{3}{2}) + b = 0 \Rightarrow b = 0$
 $f(x) = x^3 + \frac{3}{2}x^2$
 $f'(x) = 3x^2 + 3x = 3x(x+1)$
 $f'(x) = 0 \Rightarrow x = 0, -1$
 $f(0) = 1$
 $f(-1) = -1 + \frac{3}{2} = \frac{1}{2}$

⑧ $\frac{dy}{dt} = 256 - 16y$
 $\frac{dy}{dt} = 16(16 - y)$
 $\frac{dy}{16-y} = 16 dt$
 $-\ln|16-y| = 16t + C$
 $\ln|16-y| = -16t + C$
 $16-y = e^{-16t+C}$
 $16-y = e^{-16t} \cdot e^C$
 $16-y = e^{-16t} \cdot 16$
 $y = 16(1 - e^{-16t})$
 $\frac{dy}{dt} = 16 \cdot 16 e^{-16t} = 256 e^{-16t}$
 $\frac{dy}{dt} = 256 e^{-16t}$
 $\frac{dy}{dt} = 256 e^{-16t}$
 $\frac{dy}{dt} = 256 e^{-16t}$

⑨ $dx = f'(x) + f''(x) \cdot x$

$f(x) = \sin x + 2 \cos x$, for $\cos x = \frac{1}{2}$

$\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$, $\sin x = \frac{\sqrt{3}}{2}$

$\Delta y = f(x_1) - f(x_0) = \sin(\frac{\pi}{3}) + 2 \cos(\frac{\pi}{3}) - (\sin(0) + 2 \cos(0))$

$\Delta y = \frac{\sqrt{3}}{2} + 2 \cdot \frac{1}{2} - (0 + 2) = \frac{\sqrt{3}}{2} + 1 - 2 = \frac{\sqrt{3}}{2} - 1$

⑩ $f(x) = \sqrt{x+1}$, $f'(x) = \frac{1}{2\sqrt{x+1}}$

$L(x) = 1 + \frac{1}{2}x$, $\sqrt{1.1} \approx \sqrt{1.1}$

$L(0.1) = 1 + \frac{1}{2}(0.1) = 1 + \frac{1}{20} = \frac{21}{20}$

⑪ $\log_2(\cosh x)$ at $x = \ln 2$

$y' = \frac{\sinh x \cdot \frac{1}{\cosh x}}{\cosh x} = \frac{\tanh x}{\cosh x}$

$y'(\ln 2) = \frac{\tanh(\ln 2)}{\cosh(\ln 2)} = \frac{1}{\cosh(\ln 2)}$

$m_1 = \frac{1}{\cosh(\ln 2)}$

$m_2 = \frac{1}{\cosh(\ln 2)}$

$m_1 = \frac{1}{\cosh(\ln 2)}$

$m_2 = \frac{1}{\cosh(\ln 2)}$

$m_1 = \frac{1}{\cosh(\ln 2)}$

⑫ For vertical tangent we need $f'(x) = 0$

$f(x) = (3x-2)^{-1/2} = 0 \Rightarrow x = \frac{2}{3}$

⑬ $f(x) = \begin{cases} 1-x^2 & 0 \leq x \leq 2 \\ 2x-4 & 2 \leq x \leq 3 \end{cases}$

Has max at $(3, 2)$

No local max

No min or local min

⑭ $f(b) - f(a) = f'(c)(b-a)$

$f(1) - f(0) = f'(c)(1-0)$

$f(1) = 1 + e^1 = 1 + e$

$f(0) = 1 + e^0 = 1 + 1 = 2$

$f'(x) = 1 + e^x \Rightarrow 1 + e^c = e$

$e^c = e - 1$

$c = \ln(e-1)$

⑮ $f(x) = x - \ln x$

$f'(x) = 1 - \frac{1}{x} = \frac{x-1}{x}$

$f'(x) = 0 \Rightarrow x = 1$

$f(1) = 1 - \ln 1 = 1$

$f(2) = 2 - \ln 2$

$f(3) = 3 - \ln 3$

$f(4) = 4 - \ln 4$

$f(5) = 5 - \ln 5$

$f(6) = 6 - \ln 6$

$f(7) = 7 - \ln 7$

$f(8) = 8 - \ln 8$

$f(9) = 9 - \ln 9$

$f(10) = 10 - \ln 10$

$f(11) = 11 - \ln 11$

$f(12) = 12 - \ln 12$

$f(13) = 13 - \ln 13$

$f(14) = 14 - \ln 14$

$f(15) = 15 - \ln 15$

$f(16) = 16 - \ln 16$

$f(17) = 17 - \ln 17$

$f(18) = 18 - \ln 18$

$f(19) = 19 - \ln 19$

$f(20) = 20 - \ln 20$

$f(21) = 21 - \ln 21$

$f(22) = 22 - \ln 22$

$f(23) = 23 - \ln 23$

$f(24) = 24 - \ln 24$

$f(25) = 25 - \ln 25$

$f(26) = 26 - \ln 26$

$f(27) = 27 - \ln 27$

$f(28) = 28 - \ln 28$

$f(29) = 29 - \ln 29$

$f(30) = 30 - \ln 30$

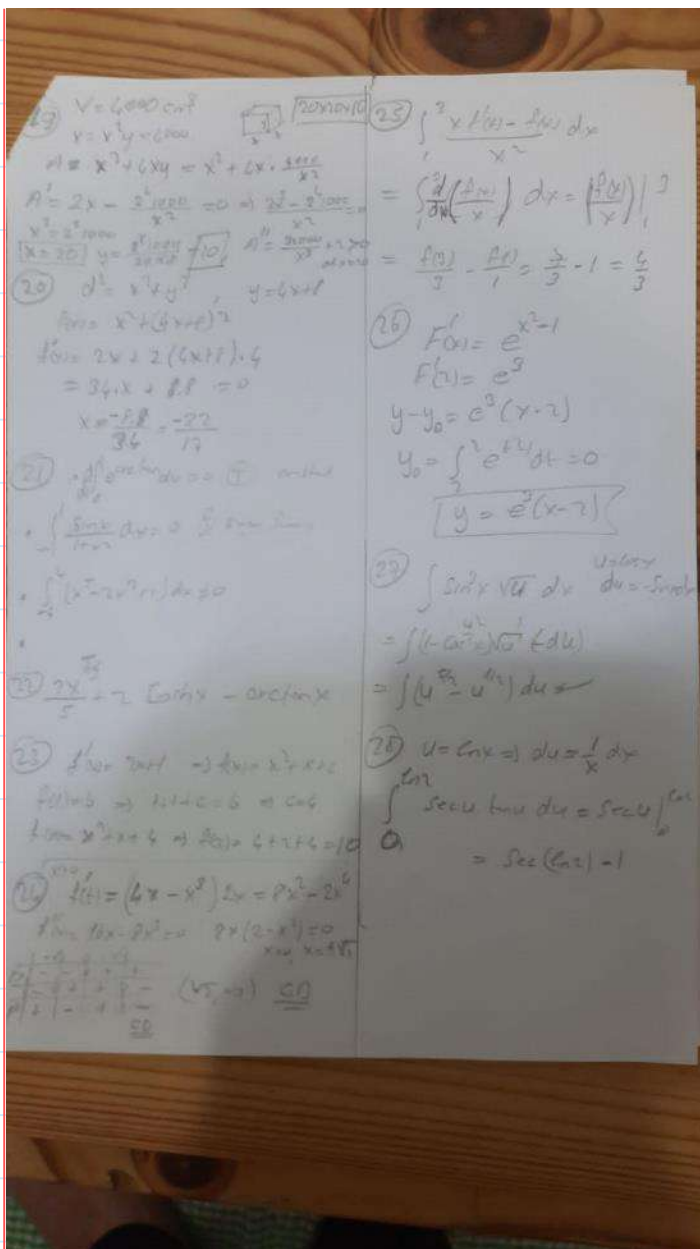
$f(31) = 31 - \ln 31$

$f(32) = 32 - \ln 32$

$f(33) = 33 - \ln 33$

$f(34) = 34 - \ln 34$

$f(35) = 35 - \ln 35$



(29) $2x^2 = x^4 - 2x^2$
 $x^4 - 4x^2 = 0$
 $x^2(x^2 - 4) = 0$
 $x^2 = 0, x^2 - 4 = 0$
 $x = 0, x = \pm 2$

$x < -2$	$-$	$+$	$-$	$+$
$-2 < x < 0$	$+$	$+$	$+$	$+$
$0 < x < 2$	$+$	$+$	$-$	$+$
$x > 2$	$+$	$+$	$+$	$+$

$A = \int_{-2}^2 (x^4 - 4x^2) dx = \int_{-2}^2 (4x^2 - x^4) dx$
 $= 2 \left(\frac{4}{3} x^3 - \frac{x^5}{5} \right) \Big|_{-2}^2 = 2 \left(\frac{128}{3} - \frac{32}{5} \right) = 2 \cdot 25 \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{176}{15}$

(30) $A = \int_0^1 \sin y dy$
 $y = \arcsin x$ or $A = 1$
 $x = \sin y$

(10) $f(x) = |x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$, $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x} = \lim_{x \rightarrow 0} 1/x = 0$

so f is diff. at $x=0$ and all x

$f'(x) = \begin{cases} 2x, & x \geq 0 \\ -2x, & x < 0 \end{cases} = 2|x|$

f'' is not def. for $x > 0$ and $x < 0$, but $\lim_{x \rightarrow 0^+} \frac{f'(x) - f'(0)}{x - 0} = \frac{2x}{x} = 2$
 $\lim_{x \rightarrow 0^-} \frac{f'(x) - f'(0)}{x - 0} = \frac{-2x}{x} = -2$

Can't find DNE, at $x=0$ but f'' changes sign from $(-)$ to $(+)$
 f has an inflection point at $x=0$

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19 October 2020

21:34

1. The value of the given limit is equal to

(1 Point)

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{3x + \sin(4x)}$$

☒ $\frac{2}{7}$ ✓

☐ $\frac{1}{2}$

☐ 1

☐ $\frac{2}{3}$

2. The value of the given limit is equal to

(1 Point)

$$\lim_{x \rightarrow 3} \frac{\ln(x+2) - \ln 5}{x-3}$$

☒ $\frac{1}{5}$ ✓

☐ $\ln 3$

☐ $\frac{1}{3}$

☐ 1

3. Question

(1 Point)

The value of $f^{(10)}\left(\frac{\pi}{3}\right)$ for $f(x) = \cos x + \sin x$ is equal to

- ☐ $\frac{1 + \sqrt{3}}{2}$
- ☐ $\frac{1 - \sqrt{3}}{2}$
- ☐ $\frac{-1 + \sqrt{3}}{2}$
- ☒ $\frac{-1 - \sqrt{3}}{2}$ ✓

4. An equation of the tangent line to the curve at the point (1,1) to the given curve is

(1 Point)

$$f(x) = \left(x^2 - 2x + \frac{1}{x} + 1\right)^3$$

- ☐ $y = -3x - 3$
- ☒ $y = -3x + 4$ ✓
- ☐ $y = 2x - 2$
- ☐ $y = -3x + 1$

5. (1 Point)

The value of $\frac{dy}{dx}$ for $2x^2y + x + y^2 = 4$ at the point (1, 1) is equal to

- ☒ $-\frac{5}{4}$ ✓
- ☐ $-\frac{2}{3}$
- ☐ $\frac{3}{7}$
- ☐ -1

6. Question

(1 Point)

The value of $f'(1)$ for $f(x) = x \tan^{-1}(2x - 1)$ is equal to

- ☐ $\frac{\pi}{4}$
- ☒ $\frac{4 + \pi}{4}$ ✓
- ☐ $-\frac{\pi}{4}$
- ☐ π

7. Each side of a cube is increasing at a rate of 2 cm/min. At what rate the volume is increasing when the side is 3 cm.

(1 Point)

- ☐ 2
- ☐ 6
- ☐ 27
- ☒ 54 ✓

8. Question

(1 Point)

The value of $\frac{dy}{dx}$ at the point $(1, 1)$ for $x^y = y^x$ is equal to

- ☐ 0
- ☒ 1 ✓
- ☐ 2
- ☐ -1

9. A particle is moving along the curve $y \ln x = 1$. As it reaches the point $(e, 1)$, the y-coordinate is decreasing at a rate of 2 cm/s. How fast is the x-coordinate of the point changing at that instant?

(1 Point)

- ☐ 1
- ☒ $2e$ ✓
- ☐ 2
- ☐ $\frac{2}{e}$

10. Question
(1 Point)

The linearization of the function $f(x) = \frac{\sqrt{x}}{x^2 + 2}$ at $x = 1$ is

- ☐ $L(x) = \frac{1}{3} - \frac{1}{18}(x - 1)$ ✓
- ☐ $L(x) = \frac{-1}{18} + \frac{1}{3}(x - 1)$
- ☐ $L(x) = \frac{1}{3} + 3(x - 1)$
- ☐ $L(x) = -18 + 2(x - 1)$

11. The value of the given limit is equal to
(1 Point)

$$\lim_{x \rightarrow \infty} \frac{e^{2x}}{5 \cosh(2x)}$$

- ☐ 1
- ☐ e
- ☐ $\frac{2}{5}$ ✓
- ☐ $\frac{2}{e}$

12. Question
(1 Point)

If $f(2) = 3$ and $f'(2) = -2$. Then the best approximate value of $f(2.03)$ is

- ☐ 3.1
- ☐ 3.0.5
- ☐ 2.94 ✓
- ☐ 2.99

13. Question
(1 Point)

If $f(x) = 2 \sin^2 x + 2 \cos x$ then one of the following numbers is a critical number of the function f .

- ☐ $\frac{5\pi}{2}$
- ☒ $\frac{7\pi}{3}$
- ☐ $\frac{13\pi}{4}$
- ☐ $\frac{5\pi}{6}$

14. Question
(1 Point)

The absolute minimum value of the function $f(x) = x^4 - x^2 + 5$ on the interval $[0, 2]$ is equal to

- ☐ 5
- ☒ $\frac{19}{4}$
- ☐ 3
- ☐ $\frac{77}{16}$

15. (1 Point)

The absolute maximum value of the function $f(x) = x - \sqrt{x}$ on the interval $[0, 3]$ is equal to

- ☐ 0
- ☐ $-\frac{1}{4}$
- ☒ $3 - \sqrt{3}$
- ☐ 6

16. The number(s) c that satisfies (satisfy) the conclusion of the Mean Value Theorem for the given function on the interval $[-3, 1]$ is (are) equal to
(1 Point)

$$f(x) = x^3 - 5x - 3$$

- ☒ -1
- ☐ 1
- ☐ 0
- ☐ ± 1

17. Question
(1 Point)

If $f(2) = 4$ and $f'(x) \geq 3$ for $2 \leq x \leq 5$. Then the smallest possible value for $f(5)$ is equal to

☒ 13 ✓

☐ 6

☐ 1

☐ 17

18. Question
(1 Point)

The function $f(x) = x^2(x^2 - 2)$ is increasing on the interval

☐ $(-\infty, -1)$

☒ $(-1, 0)$ ✓

☐ $(0, \infty)$

☐ $(0, 1)$

19. Question
(2 Points)

The function $f(x) = xe^{-x}$ is concave upward on interval

☐ $(-\infty, 2)$

☐ $(-\infty, 1)$

☒ $(2, \infty)$ ✓

☐ $(1, \infty)$

20. Question
(1 Point)

If $f(x) = (\sin x)^x$, then $f'(x) =$

☐ $x(\sin x)^{x-1}$

☐ $(\cos x)^x$

☐ $x(\sin x)^{x-1} \cos x$

☒ $(\sin x)^x (x \cot x + \ln(\sin x))$ ✓

21. (1 Point)

If f is differentiable and $g(x) = f(1 + 3x) + f(1 - 3x)$, then $g'(0) =$

- ☐ 1
- ☐ -1
- ☒ 0 ✓
- ☐ Does not exist

22. If f is continuous on its domain and has exactly three x-intercepts, then f must have
(1 Point)

- ☐ Exactly two critical numbers
- ☒ At least two critical numbers ✓
- ☐ One critical number
- ☐ No critical numbers

23. Question
(2 Points)

One of the following points is an inflection point to the graph of the function $f(x) = \frac{1}{x} + \ln x$

- ☒ $\left(2, \frac{1}{2} + \ln 2\right)$ ✓
- ☐ $(1, 1)$
- ☐ $\left(\frac{1}{3}, 3 - \ln 3\right)$
- ☐ $\left(3, \frac{1}{3} + \ln 3\right)$

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1. The value of the given limit is equal to
(1 Point)

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos(3x)}{\tan x + \sin x}$$

- ☒ $1 - \sqrt{2}$ ✓
- ☐ $\frac{3}{2}$
- ☐ 0
- ☐ -1
- ☐ None of the given choices

2. The point(s) where the given curve has horizontal tangent is (are)
(2 Points)

$$y^2 = x^2 + 2x + 4$$

- ☐ $(1, \pm\sqrt{7})$
- ☐ $(1, \pm\sqrt{7}), (-1, \pm\sqrt{3})$
- ☒ $(-1, \pm\sqrt{3})$ ✓
- ☐ $(2, \pm\sqrt{12})$
- ☐ None of the given choices

3. (1 Point)

The value of $f'(0)$ if $f(x) + x \cos(f(x)) = x^3$ is equal to

- ☒ -1 ✓
- ☐ 0
- ☐ 1
- ☐ 2
- ☐ None of the given choices

(1 Point)

The value of the second derivative of $f(x) = x^2 \ln x$ at $x = 1$ is equal to

- ☐ 1
- ☐ 0
- ☐ 2
- ☒ 3 ✓
- ☐ None of the given choices

5. Question

(1 Point)

The differential of $y = \sinh(3x) + (x + 1)^2 e^x$ at $x = 0$ with $dx = 0.01$ is equal to

- ☒ 0.06 ✓
- ☐ 0.01
- ☐ 0.03
- ☐ 0.02
- ☐ None of the given choices

6. Question

(1 Point)

The simplest form of $\tanh(\ln x)$ is

- ☒ $\frac{x^2 - 1}{x^2 + 1}$ ✓
- ☐ $\frac{2}{x^2 + 1}$
- ☐ $\frac{x^2 - 1}{2}$
- ☐ $\frac{2x}{x^2 + 1}$
- ☐ None of the given choices

7. (1 Point)

Suppose $x^5 + y^3 = 33$. If $\frac{dy}{dt} = 2$, then $\frac{dx}{dt}$ at the instant $y = 1$ is equal to

- ☐ $\frac{23}{14}$
- ☐ $\frac{-22}{13}$
- ☒ $\frac{-3}{40}$ ✓

7. (1 Point)

Suppose $x^5 + y^3 = 33$. If $\frac{dy}{dt} = 2$, then $\frac{dx}{dt}$ at the instant $y = 1$ is equal to

- ☐ $\frac{23}{14}$
- ☐ $\frac{-22}{13}$
- ☒ $\frac{-3}{40}$ ✓
- ☐ $\frac{-5}{27}$
- ☐ None of the given choices

8. Question
(1 Point)

The value of a such that the function $f(x) = ax^2 + 3x + 2$ has an absolute minimum value 1 is equal to

- ☐ 1
- ☐ $\frac{3}{2}$
- ☒ $\frac{9}{4}$ ✓
- ☐ 0
- ☐ None of the given choices

9. Question
(1 Point)

The function $f(x) = xe^{-x}$ is concave upward on interval

- ☐ $(-\infty, 2)$
- ☐ $(-\infty, 1)$
- ☒ $(2, \infty)$ ✓
- ☐ $(1, \infty)$
- ☐ None of the given choices

10. The value of the given limit is equal to

(1 Point)

$$\lim_{x \rightarrow 0} \frac{\sin((2+x)^2) - \sin 4}{x}$$

- ☐ $\sin 4$
- ☒ $4 \cos 4$ ✓
- ☐ 0
- ☐ $\frac{1}{4}$
- ☐ None of the given choices

11. Question

(1 Point)

A linear estimate of $\tan(29^\circ)$ is equal to

- ☐ $\frac{1}{\sqrt{3}} + \frac{4}{3}$
- ☐ $\frac{1}{\sqrt{3}} - \frac{4}{3}$
- ☒ $\frac{1}{\sqrt{3}} - \frac{\pi}{135}$ ✓
- ☐ $\frac{1}{\sqrt{3}} + \frac{\pi}{135}$
- ☐ None of the given choices

12. The absolute maximum and minimum values of the given function respectively are

(2 Points)

$$f(x) = \begin{cases} -4x, & \text{if } -1 \leq x < 0, \\ 1 - (x-1)^2, & \text{if } 0 \leq x \leq 3 \end{cases}$$

- ☒ $4, -3$ ✓
- ☐ $8, -3$
- ☐ $4, -8$
- ☐ $1, 0$
- ☐ None of the given choices

13. Question
(1 Point)

Let f and g be two differentiable functions such that $f' = g$ and $f = -g'$. Then $f^2 + g^2$ is equal

- ☐ x^2
- ☐ x
- ☐ 0
- ☒ Constant ✓
- ☐ None of the given choices

14. Question
(1 Point)

$$\frac{d}{dx}(\sinh^2 x - \cosh^2 x)^{2020} =$$

- ☒ 0 ✓
- ☐ $2020(\sinh^2 x - \cosh^2 x)^{2019}$
- ☐ $2020(\cosh^2 x - \sinh^2 x)(\sinh^2 x - \cosh^2 x)^{2019}$
- ☐ None of these given Choices
- ☐ 1

15. The slope of the tangent line to the curve at the given point is equal to
(1 Point)

$$x^2 + y^3 - 3y = 4, (\sqrt{2}, -1)$$

- ☐ 0
- ☐ 1
- ☒ Does not exist ✓
- ☐ None of these given choices
- ☐ -2

16. The number(s) c that satisfies (satisfy) the conclusion of the Mean Value Theorem for the given function on the given interval is (are)

(1 Point)

$$f(x) = x^3 + x^2, [-1, 1]$$

- ☐ $-1, 2$
- ☐ $1, \frac{1}{3}$
- ☐ $-1, \frac{1}{3}$
- ☒ $\frac{1}{3}$ ✓
- ☐ None of the given choices

17. (1 Point)

If f is differentiable and $g(x) = f(1 + 3x) + f(1 - 3x)$, then $g'(0) =$

- ☐ 1
- ☐ -1
- ☒ 0 ✓
- ☐ Does not exist
- ☐ None of the given choices

18. If f has exactly four x -intercepts, then f must have
- (1 Point)

- ☐ Exactly three critical numbers
- ☐ At least three critical numbers
- ☒ Nothing can be determined ✓
- ☐ No critical numbers

19. The radius of a sphere is decreasing at a rate of 2 cm/min. At what rate the volume is changing when the radius is $\frac{1}{2}$ cm?

(1 Point)

- ☐ $-\pi$
- ☐ -1
- ☐ 4π
- ☒ -2π ✓
- ☐ None of the given choices

20. An equation of the normal line to the given curve at the point (1,0) is
(1 Point)

$$f(x) = \left(x^2 - 2x + \frac{1}{x}\right)^3$$

- ☐ $y = 0$
- ☒ $x = 1$ ✓
- ☐ $y = x - 1$
- ☐ $y = -x + 1$
- ☐ None of the given choices

21. One of the following numbers is a critical number of the function the given function
(1 Point)

$$f(x) = 4x - \tan x$$

- ☐ $\frac{5\pi}{2}$
- ☒ $\frac{5\pi}{3}$ ✓
- ☐ $\frac{13\pi}{4}$
- ☐ $\frac{7\pi}{6}$
- ☐ None of the given choices

22. Question
(1 Point)

The function $f(x) = x^2 - x - \ln x$ is increasing on the interval

- ☐ $(0, 1)$
- ☒ $(1, \infty)$ ✓
- ☐ $(0, \infty)$
- ☐ $\left(\frac{-1}{2}, 1\right)$
- ☐ None of the given choices

23. The value of the given limit is equal to
(1 Point)

$$\lim_{x \rightarrow \infty} \frac{\sinh x + e^x}{7 \cosh x - 3e^x}$$

☒ 3 ✓

☐ $\frac{2}{7}$

☐ $\frac{1}{2}$

☐ $-\frac{1}{3}$

☐ None of the given choices

Final Exam Spring 2019/2020

09 July 2020 00:52

1. Question

(1 Point)

The value of the limit $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x^3 - 2x^2}$ is equal to

☒ $\frac{1}{16}$ ✓

☐ $\frac{1}{8}$

☐ $\frac{1}{32}$

☐ $\frac{3}{32}$

☐ None of the given choices

2. Question

(1.5 Points)

The domain of $f(x) = \ln\left(\frac{x+2}{x-1}\right)$ is

☒ $(-\infty, -2) \cup (1, \infty)$ ✓

☐ $(-2, 1)$

☐ $(-\infty, -1) \cup (2, \infty)$

☐ $(-1, 2)$

☐ None of the given choices

3. The value of the number b such that the given function is continuous everywhere is equal to
(1.5 Points)

$$f(x) = \begin{cases} bx^3 + 2, & \text{if } x \geq 1, \\ \tan^{-1}(e^{x-1}) + 6, & \text{if } x < 1 \end{cases}$$

☒ $\frac{16 + \pi}{4}$ ✓

☐ $\frac{12 + \pi}{4}$

☐ $\frac{8 + \pi}{4}$

☐ $\frac{4 + \pi}{4}$

☐ None of the given choices

4. Question
(1 Point)

The equation $x^6 + 6x - 23 = 0$ has at most

☐ One real root

☒ Two real roots ✓

☐ Three real roots

☐ None of the given choices

5. Question
(1.5 Points)

The equation $3x - \cos x - 17 = 0$ has

☒ Only one real root ✓

☐ Only two real roots

☐ No real roots

☐ None of the given choices

6. The value of the number b such that the given function has an infinite discontinuity at $x=b$ is equal to
(1 Point)

$$f(x) = \begin{cases} \frac{x^2 - 1}{x^2 - 3x + 2}, & \text{if } x > b, \\ x^2 - 3, & \text{if } x \leq b \end{cases}$$

- ☒ 2 ✓
- ☐ -1
- ☐ 1
- ☐ -2
- ☐ None of the given choices

7. Question
(1.5 Points)

The horizontal asymptotes of the graph of the function $f(x) = \sqrt{x^2 + 2x} - \sqrt{x^2 + 3x}$ are

- ☒ $y = \frac{\pm 1}{2}$ ✓
- ☐ $y = \pm 1$
- ☐ $y = \frac{\pm 3}{2}$
- ☐ $y = \pm 2$
- ☐ None of the given choices

8. Question
(1 Point)

The value of the limit $\lim_{x \rightarrow 0} \frac{3x}{\sin x + 5 \tan(2x)}$ is equal to

- ☒ $\frac{3}{11}$ ✓
- ☐ $\frac{5}{14}$
- ☐ $\frac{1}{5}$
- ☐ $\frac{4}{13}$
- ☐ None of the given choices

9. Question
(1.5 Points)

The area of the region enclosed by $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ is equal to

- ☒ $\frac{1}{12}$ ✓
- ☐ $\frac{1}{20}$
- ☐ $\frac{1}{10}$
- ☐ $\frac{1}{18}$
- ☐ None of the given choices

10. (1 Point)

If $\frac{d}{dx}(f(x^3)) = 6x^4$ for $x \neq 0$, then $f'(x^3)$ is equal to

- ☐ $f'(3x^2)$
- ☒ $2x^2$ ✓
- ☐ $24x^3$
- ☐ $8x$
- ☐ None of the given choices

11. Question
(1 Point)

If $\int_1^3 f(x)dx = 7$ and $\int_1^3 g(x)dx = -2$, then the value of $\int_3^1 (2f(x) - 5g(x)) dx$ is equal to

- ☒ -24 ✓
- ☐ 7
- ☐ -25
- ☐ -2
- ☐ None of the given choices

12. Question
(1.5 Points)

The value of the definite integral $\int_0^1 (3^{\frac{1}{2}} - x^3) dx$ is equal to

- ☒ $\frac{8\sqrt{3} - \ln 3 - 8}{4 \ln 3}$ ✓
- ☐ $\frac{6\sqrt{2} - \ln 2 - 6}{3 \ln 2}$
- ☐ $\frac{6\sqrt{3} - \ln 3 - 6}{3 \ln 3}$
- ☐ $\frac{8\sqrt{2} - \ln 2 - 8}{4 \ln 2}$
- ☐ None of the given choices

13. The area of a rectangle is 10 cm squared. If the length is increasing at a rate of 2 cm/min, then at what rate the width is decreasing when the length is 5 cm?
(1.5 Points)

- ☒ $\frac{4}{5}$ ✓
- ☐ $\frac{1}{5}$
- ☐ $\frac{8}{5}$
- ☐ $\frac{2}{5}$
- ☐ None of the given choices

14. Question
(1.5 Points)

The x - coordinate of the points where curve $x^2 + xy + y^2 = 3$ has a horizontal tangent are

- ☒ ± 1 ✓
- ☐ ± 2
- ☐ ± 3
- ☐ ± 4
- ☐ None of the given choices

15. Question
(1 Point)

The value of the derivative of the function $f(x) = \tanh(4 \sin^{-1}(3x))$ at $x = 0$ is equal to

- ☒ 12 ✓
- ☐ 10
- ☐ 15
- ☐ 8
- ☐ None of the given choices

16. The length L and the width W of the rectangle of the greatest area that can be inscribed in a semicircle of radius 5 metres are
(1.5 Points)

- ☐ $L = W = 5\sqrt{2}$
- ☒ $L = 5\sqrt{2}, W = \frac{5}{\sqrt{2}}$ ✓
- ☐ $L = 5, W = \sqrt{2}$
- ☐ $L = \sqrt{2}, W = 5\sqrt{2}$
- ☐ None of these given choices

17. Question
(1.5 Points)

The x - coordinate of the point on the line $y = 2x - 1$ that is closest to the point $(-1, -1)$ is equal to

- ☒ $-\frac{1}{5}$ ✓
- ☐ $\frac{1}{5}$
- ☐ $\frac{1}{2}$
- ☐ $-\frac{1}{2}$
- ☐ None of the given choices

18. Question
(1 Point)

The value of the definite integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sinh^3 x \cos x + \sin x \cosh^3 x) dx$ is equal to

- ☐ -1
- ☒ 0 ✓
- ☐ 1
- ☐ $\sqrt{3} \ln 2$
- ☐ None of the given choices

19. Question
(1.5 Points)

The value of b such that $F(x) = b \ln \left| \frac{x}{3+2x} \right|$ is an antiderivative of $f(x) = \frac{1}{3x+2x^2}$ is equal to

- ☒ $\frac{1}{3}$ ✓
- ☐ $\frac{1}{2}$
- ☐ $-\frac{1}{2}$
- ☐ $-\frac{1}{3}$
- ☐ None of these given choices

20. Question
(1.5 Points)

The value of the definite integral $\int_0^3 (x - 1 + \sqrt{9 - x^2}) dx$ is equal to

- ☒ $\frac{9\pi + 6}{4}$ ✓
- ☐ $\frac{9\pi - 6}{4}$
- ☐ $\frac{24 + 9\pi}{4}$
- ☐ $\frac{12 + 9\pi}{4}$
- ☐ None of the given choices

21. Question
(1.5 Points)

If $\int_1^3 f(x)dx = 2$ and $\int_0^1 f(x)dx = 3$ and f is an even function, then the value of $\int_{-1}^3 f(x)dx$ is equal to

- ☒ 8 ✓
- ☐ 7
- ☐ 4
- ☐ -4
- ☐ None of these given choices

22. If f is continuous, has only one x-intercept at $x=2$ with
(1 Point)

$\int_0^2 f(x)dx = 4$ and $\int_2^5 f(x)dx = -7$, then the value of $\int_0^5 |-2f(x)| dx$ is equal to

- ☒ 22 ✓
- ☐ 6
- ☐ 28
- ☐ 4
- ☐ None of these given choices

23. Question
(1.5 Points)

The value of the definite integral $\int_0^1 (8x^3 + 2)(x^4 + x)^5 dx$ is equal to

- ☒ $\frac{64}{3}$ ✓
- ☐ $\frac{32}{9}$
- ☐ $\frac{8}{3}$
- ☐ $\frac{32}{3}$
- ☐ None of the given choices

24. Question
(1.5 Points)

The value of the definite integral $\int_0^1 \frac{x^2 + 2}{x^2 + 1} dx$ is equal to

☒ $\frac{4 + \pi}{4}$ ✓

☐ $\frac{8 - \pi}{4}$

☐ $\frac{-4 + 3\pi}{4}$

☐ $\frac{4 - 3\pi}{4}$

☐ None of the given choices

25. Question
(1.5 Points)

The area of the region enclosed by $y = x^2$ and $y = 1$ is equal to

☒ $\frac{4}{3}$ ✓

☐ $\frac{32}{3}$

☐ $\frac{8}{3}$

☐ $\frac{16}{3}$

☐ None of the given choices

26. The x-coordinate of the point where the function
(1.5 Points)

$f(x) = \int_1^x \frac{t^2 - 5t + 4}{t^2 + 3t + 3} dt$ has a local minimum value is equal to

- ☒ 4 ✓
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ None of the given choices

27. Question
(1.5 Points)

The graph of the function $f(x) = \int_2^x \frac{t^2}{t^2 + 2} dt$ is concave upward on the interval

- ☒ (0, 2) ✓
- ☐ (2, ∞)
- ☐ $(0, \frac{1}{2})$
- ☐ $(\frac{1}{2}, \infty)$
- ☐ None of the given choices

28. Question
(1 Point)

The general indefinite integral of $\int \frac{\sinh(\sqrt{x})}{\sqrt{x}} dx$ is equal to

- ☒ $2 \cosh(\sqrt{x}) + C$ ✓
- ☐ $-2 \cosh(\sqrt{x}) + C$
- ☐ $2 \sinh(\sqrt{x}) + C$
- ☐ $-2 \sinh(\sqrt{x}) + C$
- ☐ None of the given choices

29. Question
(1 Point)

The general indefinite integral $\int \frac{4e^x}{(1 + 2e^x)^3} dx$ is equal to

- ☐ $\frac{-1}{(1 + 2e^x)^2} + C$ ✓
- ☐ $\frac{-1}{2(1 + 2e^x)^2} + C$
- ☐ $\frac{-1}{4(1 + 2e^x)^2} + C$
- ☐ $\frac{-2}{(1 + 2e^x)^2} + C$
- ☐ None of the given choices

30. Question
(1 Point)

Given that $f''(x) = 2 + e^x$ and the slope of the tangent line at the point $(0, 2)$ is 2, then

- ☐ $f(x) = e^x + x^2 + 1$
- ☐ $f(x) = e^x + x^2 + x + 2$
- ☐ $f(x) = e^x + x^2 + x + 1$ ✓
- ☐ $f(x) = e^x + x^2 + C$
- ☐ None of the given choices

31. Question
(1 Point)

If $f(x) = x^2 \sin x$, then $\int f'(x) dx$ is equal to

- ☐ $\frac{-x^3}{3} \cos x + C$
- ☐ $2x \sin x + x^2 \cos x + C$
- ☐ $x^2 \sin x + C$ ✓
- ☐ $2x \cos x + C$
- ☐ None of the given choices

Final Exam fall 2021 10_3_2021

12 March 2021 20:18

1. Question

(3 Points)

The domain of the function $f(x) = \log_3\left(\frac{x-1}{x+2}\right)$ is

- ☐ $(-2, 1)$
- ☐ $(-\infty, 0) \cup (2, \infty)$
- ☐ $(1, \infty)$
- ☒ $(-\infty, -2) \cup (1, \infty)$ ✓

2. Question

(3 Points)

If $\tan^{-1} x = \sqrt{y}$, then $\sin(2\sqrt{y})$ is equal to

- ☐ 0
- ☐ $\frac{1}{\sqrt{x}}$
- ☐ $2\sqrt{x}$
- ☒ $\frac{2x}{1+x^2}$ ✓
- ☐ $2 + 2x$

3. Question
(3 Points)

The value of $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan(3x)}{x}$ is equal to

- ☐ 3
- ☐ 0
- ☒ $\frac{-4}{\pi}$ ✓
- ☐ $\tan 3$
- ☐ π

4. Question
(4 Points)

One of the following intervals contains a root of the function $f(x) = \ln x - x + 3$:

- ☐ (1, 2)
- ☒ (1, e^2) ✓
- ☐ (e^2 , e^3)
- ☐ (e^{-1} , e^{-2})
- ☐ None of the given choices

5. The value of the number a such that f is continuous at $x=1$ where
(3 Points)

$f(x) = \begin{cases} 3ax^2 + 3x, & \text{if } x < 1, \\ x^3 - a^2x, & \text{if } x \geq 1 \end{cases}$ is equal to

- ☐ -3
- ☐ 2
- ☐ 1
- ☒ -1 ✓
- ☐ 4

6. Question
(3 Points)

The value of $\lim_{x \rightarrow \infty} \frac{2e^{x+1} - 3}{e^x + 1}$ is equal to

- ☐ $-3e$
- ☒ $2e$ ✓
- ☐ e
- ☐ $-e$
- ☐ 0

7. Question
(3 Points)

The value of $\lim_{x \rightarrow 1} \frac{x^{100} - 1}{x - 1}$ is equal to

- ☐ 0
- ☐ 1
- ☒ 100 ✓
- ☐ 99
- ☐ Does not exist

8. Question
(4 Points)

The slope of the normal line to the curve $y = xe^{x^2}$ at $x = 1$ is equal to

- ☐ $2e$
- ☐ $\frac{-1}{e}$
- ☒ $\frac{-1}{3e}$ ✓
- ☐ -1
- ☐ None of these given choices

9. Question
(4 Points)

A value of the number c such that $3y'' + 2y' - y = 0$ for $y = e^{cx}$ is equal to

- ☐ $-\frac{1}{6}$
- ☐ 1
- ☒ $\frac{1}{3}$ ✓
- ☐ $-\frac{1}{2}$
- ☐ 0

10. Question
(4 Points)

The value of $\frac{dy}{dx}$ at $x = 0$ for the curve $x^2 + xy + y^3 = 1$ is equal to

- ☐ 4
- ☐ 2
- ☒ $-\frac{1}{3}$ ✓
- ☐ 0
- ☐ $\frac{1}{2}$

11. Question
(4 Points)

The absolute maximum value of the function $f(x) = \frac{\sqrt{x}}{3+x^2}$ on the interval $[0, 4]$ is equal to

- ☐ 0
- ☐ 1
- ☐ $\frac{2}{19}$
- ☒ $\frac{1}{4}$ ✓
- ☐ $\frac{2}{5}$

12. Question
(3 Points)

The number c that satisfies the conclusion of Rolle's Theorem for $f(x) = x + \frac{1}{x}$ on $\left[\frac{1}{3}, 3\right]$ is equal to

- ☒ 1 ✓
- ☐ -1
- ☐ 0
- ☐ $\frac{10}{3}$
- ☐ $\frac{1}{4}$

13. Question
(3 Points)

The maximum area of a rectangle of perimeter 10 meters in square meters is equal to

- ☒ $\frac{25}{4}$ ✓
- ☐ 6
- ☐ 4
- ☐ $\frac{100}{8}$
- ☐ $\frac{15}{2}$

14. Question
(4 Points)

The minimum vertical distance between the curves $y = x^2 + 1$ and $y = x$ is equal to

- ☒ $\frac{3}{4}$ ✓
- ☐ 1
- ☐ $\frac{4}{7}$
- ☐ $\frac{1}{4}$
- ☐ $\frac{3}{2}$

15. Question
(3 Points)

The function f such that $f'(x) = 3x^2 - 5$ and the line $y + 2x = 0$ is tangent to the graph of f is

- ☐ $f(x) = x^3 - 5x + 4$
- ☐ $f(x) = 6x$
- ☒ $f(x) = x^3 - 5x - 2$ ✓
- ☐ $f(x) = 6x + 2$
- ☐ None of these given choices

16. Suppose that the function $y = f(x)$ is continuous on $\mathbb{R}$ and it is an odd function.

(3 Points)

If $\int_2^5 f(x)dx = -4$ and, then the value of $\int_5^{-2} f(x)dx$ is equal to

- ☒ 4 ✓
- ☐ -4
- ☐ 2
- ☐ 0
- ☐ 20

17. Question

(3 Points)

If $\int_1^3 f(x)dx = 2$ and $\int_1^3 g(x)dx = 7$, then the value of $\int_3^1 (2f(x) - 3g(x))dx$ is equal to

- ☒ 17 ✓
- ☐ 8
- ☐ -8
- ☐ -17
- ☐ 25

18. Question

(4 Points)

An interval on which the function $f(x) = \int_0^x (t^2 - 3t + 2)e^t dt$ is decreasing is

- ☐ $(-\infty, 1)$
- ☐ $(2, \infty)$
- ☒ $(1, 2)$ ✓
- ☐ $(-\infty, 2)$
- ☐ $(1, \infty)$

19. Question
(4 Points)

One of the following is a local maximum value of the function $f(x) = \int_1^x \frac{t^2 - 1}{t^2 + 1} dt$

- ☐ -2
- ☐ 0
- ☒ $-2 + \pi$ ✓
- ☐ $\frac{\pi}{4}$
- ☐ -1

20. Question
(3 Points)

If $f(2) = 5$, f' is continuous and $\int_2^4 f'(x) dx = 12$, then the value of $f(4)$ is equal to

- ☐ 12
- ☐ 7
- ☐ 0
- ☐ 1
- ☒ 17 ✓

21. Question
(3 Points)

The general indefinite integral of $\int \sec x dx$ is equal to

- ☐ $\sec x \tan x + C$
- ☒ $\ln |\sec x + \tan x| + C$ ✓
- ☐ $\cos x + C$
- ☐ $-\csc x + C$
- ☐ None of these given choices

22. Question
(4 Points)

The value of the definite integral $\int_0^3 |2x - 4| dx$ is equal to

- ☐ 2
- ☒ 5 ✓
- ☐ 4
- ☐ 3
- ☐ 0

23. Question
(3 Points)

The general indefinite integral of $\int \cos x \sin(\sin x) dx$ is equal to

- ☐ $\sin x \cos(\cos x) + C$
- ☐ $\sin x \cos(\sin x) + C$
- ☒ $-\cos(\sin x) + C$ ✓
- ☐ $\sin(\sin x) + C$
- ☐ None of these given choices

24. Question
(4 Points)

The value of the definite integral $\int_0^1 \frac{x}{1+x^4} dx$ is equal to

- ☒ $\frac{\pi}{8}$ ✓
- ☐ 0
- ☐ $\frac{\pi}{2}$
- ☐ π
- ☐ 1

25. Question
(3 Points)

The area of the region under the curve $y = 3^x$ from $x = 0$ to $x = \log_3 4$ is equal to

- ☐ $\frac{1}{\ln 3}$
- ☒ $\frac{3}{\ln 3}$ ✓
- ☐ $2 \ln 3$
- ☐ $4 \ln 3$
- ☐ 1

26. Question
(4 Points)

The area of the region enclosed by the curves $y = x^2$ and $y = 3 - 2x$ is equal to

- ☒ $\frac{32}{3}$ ✓
- ☐ $\frac{15}{2}$
- ☐ $\frac{25}{4}$
- ☐ $\frac{8}{5}$
- ☐ None of these given choices

27. Question
(4 Points)

The area of the region enclosed by the curves $y = 3 + x$ and $y = 3 - x$ and the x -axis is equal to

- ☒ 9 ✓
- ☐ 6
- ☐ 3
- ☐ 12
- ☐ None of these given choices

28. Question
(3 Points)

The derivative of $F(x) = \int_x^{x^2} e^{t^2+1} dt$ at $x = 1$ is equal to

- ☐ e
- ☒ e^2 ✓
- ☐ e^4
- ☐ $3e^2$
- ☐ 2

29. Question
(4 Points)

An inflection point of the function $F(x) = \int_0^x \frac{t}{1+t^2} dt$ is

- ☐ $(0, 0)$
- ☒ $(-1, \ln \sqrt{2})$ ✓
- ☐ $(\sqrt{3}, 0)$
- ☐ $(0, -\sqrt{3})$
- ☐ $(1, 1)$

1st Exam Spring 2021

07 July 2021 17:24

1. [15 pts.] Find the domain, range and the formula for the inverse of the function $f(x) = e^{2x+4}$.

$$4(x-2)$$

$$y = \left(\frac{1}{2} + \sqrt{2x+4}\right)$$

2. [15 pts.] Find the value of $\lim_{x \rightarrow 2} \frac{\sqrt{4x+1}-3}{x^2-x-6}$ if it exists.

$$\ln y = 2 + \sqrt{2x+4}$$

3. [15 pts.] Find the horizontal and vertical asymptotes of the function $f(x) = \frac{4x-1}{x(x-1)}$.

$$\ln y - 2 = \sqrt{2x+4}$$

$$(\ln y - 2)^2 = 2x+4$$

4. [15 pts.] Find and classify the points of discontinuity of the function

$$f(x) = \begin{cases} \frac{1}{x+2}, & \text{if } x < -2 \\ \ln(3x+1), & \text{if } -2 \leq x \leq 3 \\ \frac{x^2-1}{x-1}, & \text{if } x > 3 \\ 2, & \text{if } x = 1 \end{cases}$$

$$\frac{(\ln y - 2)^2}{2} = 4 = \frac{2x}{2}$$

5. [15 pts.] Use the Intermediate Value Theorem to show that the equation $2 \tan^{-1} x - 2 = x$ has at least one real root.

6. [15 pts.] Find an equation of the tangent line to the graph of $f(x) = \frac{2 \tan x}{x^2+1}$ at $x = 0$.

7. [10 pts.] Find the points on the graph of $f(x) = \frac{x^2-3}{x+2}$ where the tangent line is horizontal.

$$f^{-1}(x) = \frac{(\ln y - 2)^2 - 4}{2}$$

Df: $\ln$

$$x > 0$$

$$(0, \infty)$$

$$\ln(x-2)$$

$$x-2 > 0$$

$$x > 2$$

$$5) 2 \tan^{-1} x - 2 + x = 0$$

$$R(x) = 2 \tan^{-1} x - 2 + x$$

$$f(0) = 2 \tan^{-1} 0 - 2 + 0 = -2 < 0$$

$$f(1) = 2 \tan^{-1} 1 - 2 + 1 = \frac{\pi}{2} - 1 > 0$$

$$f\left(\frac{\pi}{2}\right) = 2 \tan^{-1} \left(\frac{\pi}{2}\right) - 2 + \frac{\pi}{2} = \frac{\pi}{2} - 1 > 0$$

$$= \frac{\pi}{2} - 1 > 0$$

$$= \frac{\pi}{2} - 1 > 0$$

$$Df^{-1} = Rf = (0, \infty)$$

$$Df = R_{f^{-1}}$$

$$2x+4 > 0$$

$$2x > -4$$

$$x > -2$$

First
Midterm ...

1. [5 + 5 + 5 + 5 + 5 = 25 pts.] The graph of the function f is given. Use it to answer each of the following:

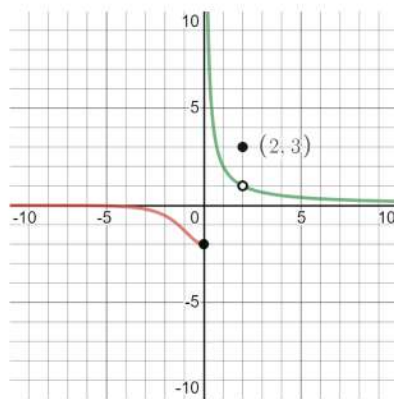


Figure 1: The graph of the function $y = f(x)$.

- (a) Find $\lim_{x \rightarrow 2} f(x)$, if it exists. Justify your answer.

$$\lim_{x \rightarrow 2} f(x) = 1$$

- (b) Find $\lim_{x \rightarrow 0} f(x)$, if it exists. Justify your answer.

$$\lim_{x \rightarrow 0} f(x) = \text{Does not exist because } \lim_{x \rightarrow 0^+} f(x) = \infty.$$

- (c) Find all vertical and horizontal asymptotes.

$x = 0$ is the only vertical asymptote and $y = 0$ is the only horizontal asymptote.

- (d) Is f continuous at $x = 0$? Explain why or why not?

No f is not continuous at $x = 0$ because $\lim_{x \rightarrow 0} f(x) = \text{Does not exist}$.

- (e) Is f differentiable at $x = 2$? Explain why or why not?

No f is not differentiable at $x = 2$ because it is not continuous at $x = 2$.

2. [2 + 5 + 3 = 10 pts.] Let $f(x) = e^{\frac{2x}{x+1}}$.

- (a) Find the domain of f .

- (b) Find a formula for the inverse of f .

- (c) Find the range of f .

The domain of f is $\mathbb{R} \setminus \{-1\}$. We have $f(x) = e^{\frac{2x}{x+1}}$ for all $x \in \mathbb{R} \setminus \{-1\}$.

(c) Find the range of f .

The domain of f is $\mathbb{R} \setminus \{-1\}$. Now we let $y = e^{\frac{2x}{x+1}}$. We apply the natural logarithm to both sides to obtain $\ln y = \frac{2x}{x+1}$. If we solve for x in terms of y , we obtain $x = \frac{\ln y}{2 - \ln y}$. Thus the inverse of f is given by $f^{-1}(x) = \frac{\ln x}{2 - \ln x}$. Finally the range of f is equal to the domain of the inverse of f that is $(0, \infty) \setminus \{e^2\}$.

3. [10 pts.] Find the exact value of $\sin\left(\cos^{-1}\left(\frac{2}{3}\right)\right)$.

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We let $x = \cos^{-1}\left(\frac{2}{3}\right)$ with $0 \leq x \leq \pi$, hence $\sin x \geq 0$. Now $\cos x = \frac{2}{3}$. In this case, we have

$$\sin\left(\cos^{-1}\left(\frac{2}{3}\right)\right) = \sin x = \sqrt{1 - \cos^2 x} = \sqrt{1 - \frac{4}{9}} = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$$

4. [5 + 5 = 10 pts.] Find the value of the limit, if it exists.

(a) $\lim_{x \rightarrow 2} \frac{\sqrt{4x+1} - 3}{x - 2}$

(b) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2}$

(a) We have

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\sqrt{4x+1} - 3}{x - 2} &= \lim_{x \rightarrow 2} \frac{\sqrt{4x+1} - 3}{x - 2} \times \frac{\sqrt{4x+1} + 3}{\sqrt{4x+1} + 3} \\ &= \lim_{x \rightarrow 2} \frac{4x + 1 - 9}{(x - 2)\sqrt{4x+1} + 3} = \lim_{x \rightarrow 2} \frac{4}{\sqrt{4x+1} + 3} = \frac{2}{3} \end{aligned}$$

(b) We have

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(x-2)} = \lim_{x \rightarrow 1} \frac{(x+1)}{(x-2)} = -2$$

5. [15 pts.] Find and classify the points of discontinuity of the function

$$f(x) = \begin{cases} 2^x, & \text{if } x \leq 0, \\ \frac{x^2 - 4}{x^2 + x - 2}, & \text{if } 0 < x < 1 \\ \frac{\sqrt{3x+3}}{x+3}, & \text{if } x \geq 1. \end{cases}$$

We have $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 2^x = 1$, and $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x^2 - 4}{x^2 + x - 2} = 2$. Therefore, f

We have $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 2^x = 1$, and $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x^2 - 4}{x^2 + x - 2} = 2$. Therefore, f is discontinuous at $x = 0$ with jump discontinuity. Also, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{x^2 - 4}{x^2 + x - 2} = \lim_{x \rightarrow 1^-} \frac{x^2 - 4}{(x + 2)(x - 1)} = \infty$. Thus f is discontinuous at $x = 1$ with infinite discontinuity. Finally, f is defined everywhere else in terms of exponential, rational and root functions. Therefore, by theorem, f is continuous on $\mathbb{R} \setminus \{0, 1\}$.

6. [10 pts.] Use the Intermediate Value Theorem to show that the equation $e^x - 3 = x^3 - 3x$ has at least one real root.

We define $f(x) = e^x - x^3 + 3x - 3$ on $[0, 1]$. Then it is easy to see that f is continuous on $[0, 1]$ since it is a difference of two continuous functions an exponential function and a polynomial. Moreover, we have $f(0) = -2 < 0 < e - 1 = f(1)$. Now using the Intermediate Value Theorem, there is at least one c in $(0, 1)$ such that $f(c) = 0$ which implies that $e^c - c^3 + 3c - 3 = 0$ or equivalently $e^c - 3 = c^3 - 3c$. Thus c is a root of the above equation.

Page 2 of 3

7. [10 pts.] Use the definition of the derivative to find $f'(2)$ for $f(x) = \frac{x}{x+3}$.

$$f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = \lim_{x \rightarrow 2} \frac{\frac{x}{x+3} - \frac{2}{5}}{x - 2} = \lim_{x \rightarrow 2} \frac{5x - 2x - 6}{5(x+3)(x-2)} = \lim_{x \rightarrow 2} \frac{3}{5(x+3)} = \frac{3}{25}$$

8. [10 pts.] Find an equation of the tangent line to the graph of $f(x) = \frac{2x+1}{x^2 e^x}$ at $x = 1$.

We have $f'(x) = \frac{2(x^2 e^x) - (2x+1)(x^2 e^x + 2x e^x)}{(x^2 e^x)^2}$. Thus $f'(1) = \frac{2e - 9e}{e^2} = \frac{-7}{e}$. Now $f(1) = \frac{3}{e}$. Therefore, an equation of the tangent line is $y - \frac{3}{e} = \frac{-7}{e}(x - 1)$.

2nd Exam 2021

07 July 2021 17:25

1. [15 pts.] Find an equation of the normal line to the curve $f(x) = \frac{2 \cosh x}{1 + \ln(x^2 + x + 1)}$ at $x = 0$.
2. [15 pts.] Show that $\frac{d^2y}{dx^2} = \frac{-2x}{y^5}$, if $x^3 + y^3 = 1$.
3. [15 pts.] Gas is escaping from a spherical balloon at a rate of $12\text{m}^3/\text{min}$. How fast is the radius decreasing volume is $4\sqrt{3}\pi \text{ m}^3$?
4. [15 pts.] Find all critical numbers and classify each as local minimum, local maximum, absolute minimum, absolute maximum, or none for the function $f(x) = x^2e^{-x}$ on $[-1, 2]$.
5. [25 pts.] Find the intervals on which the graph of f is increasing or decreasing, the intervals on which the graph of f is concave up or down and then sketch the graph of the function $f(x) = \sin x + \cos x$ on $[0, 2\pi]$.
6. [15 pts.] Show that $f(x) = x^5 + x^3 + 3x^2$ has at most one inflection point.

Final Exam Spring 2021

07 July 2021 17:25

1. Question (4 Points)

The value of $\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right)$ is equal to

- ☒ $\frac{-1}{2}$ ✓
- ☐ 0
- ☐ 1
- ☐ $\frac{2}{3}$
- ☐ None of the given choices

2. Question (4 Points)

The range of the function $f(x) = \frac{x-1}{2x+3}$ is

- ☒ $\left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$ ✓
- ☐ $\left(-\infty, \frac{-3}{2}\right) \cup \left(\frac{-3}{2}, \infty\right)$
- ☐ $\mathbb{R}$
- ☐ $\left(-\infty, \frac{-1}{3}\right) \cup \left(\frac{-1}{3}, \infty\right)$
- ☐ None of the given choices

3. The given function at $x=1$ has
(4 Points)

$$f(x) = \begin{cases} \frac{x^2 - 1}{x^3 + 1}, & \text{if } x \leq 1, \\ \frac{\ln(x - 1)}{x^2 + 1}, & \text{if } x > 1 \end{cases}$$

- ☒ infinite discontinuity ✓
- ☐ removable discontinuity
- ☐ jump discontinuity
- ☐ horizontal asymptote
- ☐ None of the given choices

4. One of the following lines is a horizontal asymptote to the graph of the given function
(4 Points)

$$f(x) = \frac{2x^2 \tan^{-1} x + 5x}{1 + x^2}.$$

- ☐ $y = 2$
- ☒ $y = \pi$ ✓
- ☐ $y = 0$
- ☐ $y = 5$
- ☐ None of the given choices

5. Question
(3 Points)

The value of $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(3x)}{4x}$ is equal to

- ☐ $\frac{3}{4}$
- ☐ $\frac{4}{3}$
- ☒ $\frac{-1}{2\pi}$ ✓
- ☐ 0
- ☐ None of the given choices

6. Question
(3 Points)

If $f(x) = \ln(3 + (g(x))^2)$ with $g(1) = 2$ and $g'(1) = 4$, then $f'(1)$ is equal to

- ☐ $\ln 7$
- ☒ $\frac{16}{7}$ ✓
- ☐ $\frac{8}{5}$
- ☐ $\frac{7}{8}$
- ☐ None of the given choices

7. Question
(3 Points)

If $\lim_{x \rightarrow 1} (f(x) + g(x)) = 4$ and $\lim_{x \rightarrow 1} (f(x) - g(x)) = 6$, then $\lim_{x \rightarrow 1} (f(x)g(x))$ is equal to

- ☒ -5 ✓
- ☐ 10
- ☐ 24
- ☐ -2
- ☐ None of the given choices

8. Question
(4 Points)

The absolute maximum of the function $f(x) = \frac{\ln x}{x}$ on $[1, e^2]$ is equal to

- ☒ $\frac{1}{e}$ ✓
- ☐ $\frac{2}{e^2}$
- ☐ $\frac{3}{e}$
- ☐ 1
- ☐ None of the given choices

9. Question
(3 Points)

The value of $\lim_{x \rightarrow \infty} \frac{2 \sinh x + 3 \cosh x}{3e^x - 2}$ is equal to

- ☐ $\frac{2}{3}$
- ☐ 1
- ☒ $\frac{5}{6}$ ✓
- ☐ 0
- ☐ None of the given choices

10. Question
(3 Points)

The slope of the normal line to the curve $x^2 + 3xy + y^2 = 2$ at $x = 0$ is equal to

- ☐ $-\frac{3}{2}$
- ☒ $\frac{2}{3}$ ✓
- ☐ 1
- ☐ $\frac{3}{5}$
- ☐ None of the given choices

11. The number c that satisfies the conclusion of the Mean Value Theorem
(3 Points)

for the function $f(x) = x + \frac{1}{x}$ on $[1, 3]$ is equal to

- ☐ 2
- ☐ $\sqrt{2}$
- ☒ $\sqrt{3}$ ✓
- ☐ $\frac{1}{\sqrt{2}}$
- ☐ None of the given choices

12. Question

(3 Points)

If $f(2) = 5$ and $f'(x) \geq 2$ for all values of x . Then the smallest possible value of $f(4)$ is

- ☐ 5
- ☒ 9 ✓
- ☐ 10
- ☐ 0
- ☐ None of the given choices

13. Question

(4 Points)

If $f'(x) = x^3$ and the line $x + y = 0$ is tangent to the graph of f , then $f(x)$ is equal to

- ☐ $\frac{x^4}{4} + 1$
- ☒ $\frac{x^4}{4} + \frac{3}{4}$ ✓
- ☐ $\frac{x^4}{4} + \frac{1}{3}$
- ☐ $\frac{x^4}{4} + \frac{5}{3}$
- ☐ None of the given choices

14. The two numbers whose difference is 10 and whose product is minimum are

(4 Points)

- ☐ 15, 5
- ☐ 3, -7
- ☐ 4, -6
- ☐ 10, 0
- ☒ None of the given choices ✓

15. Question

(4 Points)

The closest point on the curve $y = \sqrt{x}$ to the point $(1, 0)$ is

- ☒ $\left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ ✓
- ☐ $(1, 1)$
- ☐ $(1, 0)$
- ☐ $\left(\frac{3}{4}, \frac{1}{2}\right)$
- ☐ None of the given choices

16. If f is not continuous at $x=a$, then

(3 Points)

- ☒ f is not differentiable at $x = a$ ✓
- ☐ $\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$
- ☐ $f(a)$ is not defined
- ☐ $f(a) = \lim_{x \rightarrow a} f(x)$
- ☐ None of the given choices

17. Question

(4 Points)

The area of the region enclosed by $y = 6 - x^2$ and $y = x^2 - 2$ is equal to

- ☒ $\frac{64}{3}$ ✓
- ☐ $\frac{17}{2}$
- ☐ $\frac{32}{7}$
- ☐ $\frac{16}{5}$
- ☐ None of the given choices

18. Question
(4 Points)

The function $f(x) = \int_1^{x^2} \frac{dt}{t^2 + 1}$ at $x = 0$ has a

- ☐ local maximum value
- ☒ local minimum value ✓
- ☐ vertical tangent
- ☐ absolute maximum value
- ☐ None of the given choices

19. Question
(4 Points)

The function $f(x) = \int_1^x \sqrt{t^2 + 1} dt$ is

- ☒ concave up on $\left(-\frac{1}{2}, \infty\right)$ ✓
- ☐ concave down on $(0, \infty)$
- ☐ concave up on $(-\infty, 0)$
- ☐ concave up on $\left(-1, -\frac{1}{2}\right)$
- ☐ None of the given choices

20. If water is being poured in, at a rate of 3 cubic meter per minute, the water level will rise in a cylindrical can of radius 2 meter at rate equal to
(4 Points)

- ☒ $\frac{3}{4\pi} m/min$ ✓
- ☐ $3m/min$
- ☐ $\frac{4\pi}{5} m/min$
- ☐ $\frac{4}{7\pi} m/min$
- ☐ None of the given choices

21. Question
(3 Points)

The value of the definite integral $f(x) = \int_1^0 \sqrt{1-x^2} dx$ is equal to

- ☒ $-\frac{\pi}{4}$ ✓
- ☐ $-\frac{\pi}{2}$
- ☐ $-\pi$
- ☐ 0
- ☐ None of the given choices

22. Question
(3 Points)

The value of the indefinite integral $\int \frac{e^{\sqrt{x}} \cos(e^{\sqrt{x}})}{\sqrt{x}} dx$ is equal to

- ☒ $2 \sin(e^{\sqrt{x}}) + C$ ✓
- ☐ $e^{\sqrt{x}} \sin(e^{\sqrt{x}}) + C$
- ☐ $\sin(\sqrt{x}) + C$
- ☐ $\cos(e^{\sqrt{x}}) + C$
- ☐ None of the given choices

23. Question
(3 Points)

The value of the indefinite integral $\int \frac{dx}{x \ln x}$ is equal to

- ☒ $\ln |\ln x| + C$ ✓
- ☐ $\ln |x| + C$
- ☐ $\frac{1}{x} + C$
- ☐ $\ln \left| \frac{1}{x} \right| + C$
- ☐ None of the given choices

24. The value of the Riemann sum for
(3 Points)

$f(x) = x^2, 0 \leq x \leq 1$ with $n = 2$ taking the sample points to be the midpoints is equal to

- ☒ $\frac{5}{16}$ ✓
- ☐ $\frac{1}{4}$
- ☐ $\frac{5}{8}$
- ☐ $\frac{1}{4}$
- ☐ None of the given choices

25. Question
(3 Points)

The value of the definite integral $f(x) = \int_1^e \frac{dx}{x^4(\ln x)^3}$ is equal to

- ☐ 0
- ☐ $\frac{1}{3}$
- ☒ 4 ✓
- ☐ $\frac{3}{4}$
- ☐ None of the given choices

26. Question
(3 Points)

The value of the indefinite integral $\int \operatorname{sech}(\sinh x) \tanh(\sinh x) \cosh x dx$ is equal to

- ☒ $-\operatorname{sech}(\sinh x) + C$ ✓
- ☐ $\operatorname{sech}(\sinh x) + C$
- ☐ $\sinh(\sinh x) + C$
- ☐ $-\frac{\operatorname{sech} x \sinh^2 x}{2} + C$
- ☐ None of the given choices

27. If f is an odd continuous function and
(3 Points)

$\int_3^5 f(x)dx = 4$, then the value of $\int_{-3}^5 (2f(x) + 1)dx$ is equal to

- ☐ 4
- ☐ 8
- ☐ 12
- ☐ -8
- ☐ None of the given choices ✓

28. The area of the region enclosed by curves
(4 Points)

$y = \sqrt[4]{(2x-1)^3}$, $y = 0$ and $x = 1$ is equal to

- ☐ $\frac{2}{7}$ ✓
- ☐ $\frac{4}{7}$
- ☐ $\frac{1}{7}$
- ☐ 1
- ☐ None of the given choices

29. Question
(3 Points)

If $f(2) = 3$ and $f'(2) = -4$, then the linear approximation of $f(2.05)$ is equal to

- ☐ 2.8 ✓
- ☐ 2.95
- ☐ 3.05
- ☐ 2.99
- ☐ None of the given choices